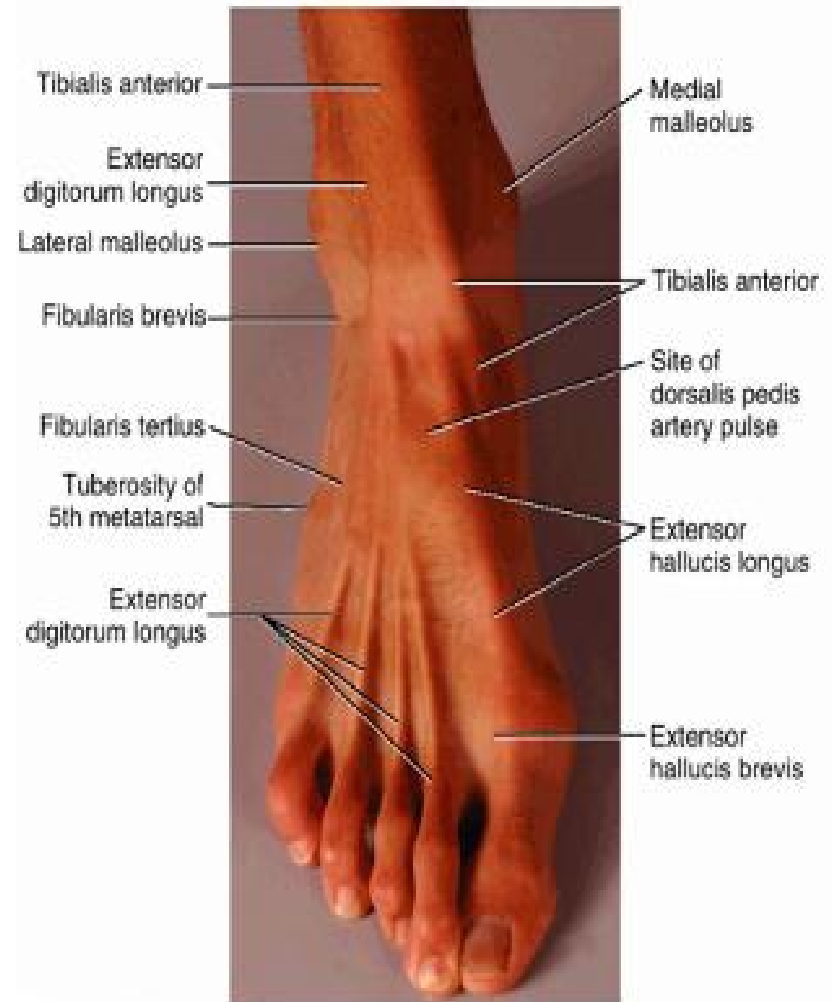


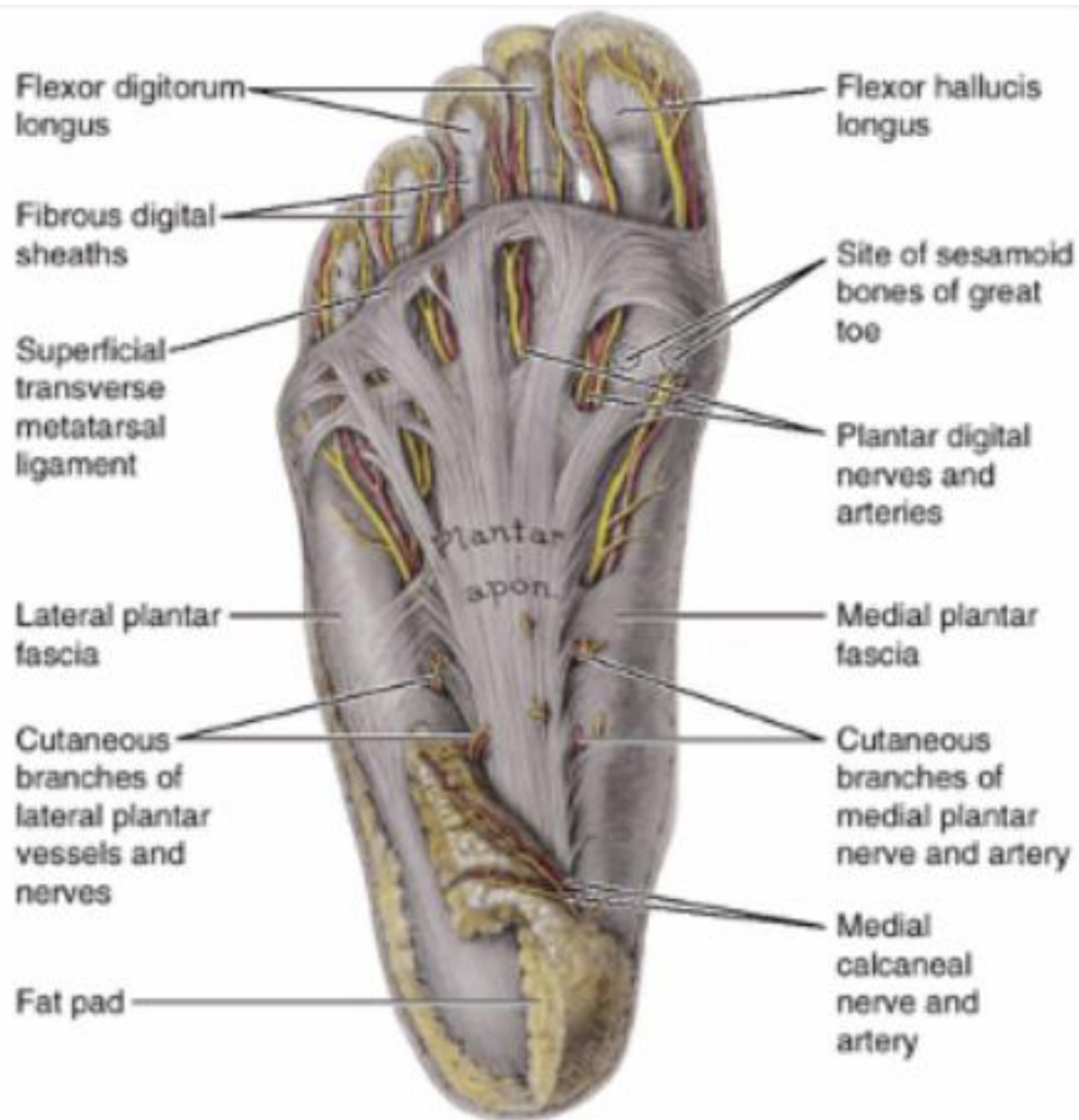
THE FOOT

- The foot is the region of the lower limb distal to the ankle joint
- provides a platform for supporting the weight of the body when standing.
- has an important role in locomotion.
- The foot has a superior surface (**dorsum of foot**) and an inferior surface (**sole**)



Deep fascia of foot

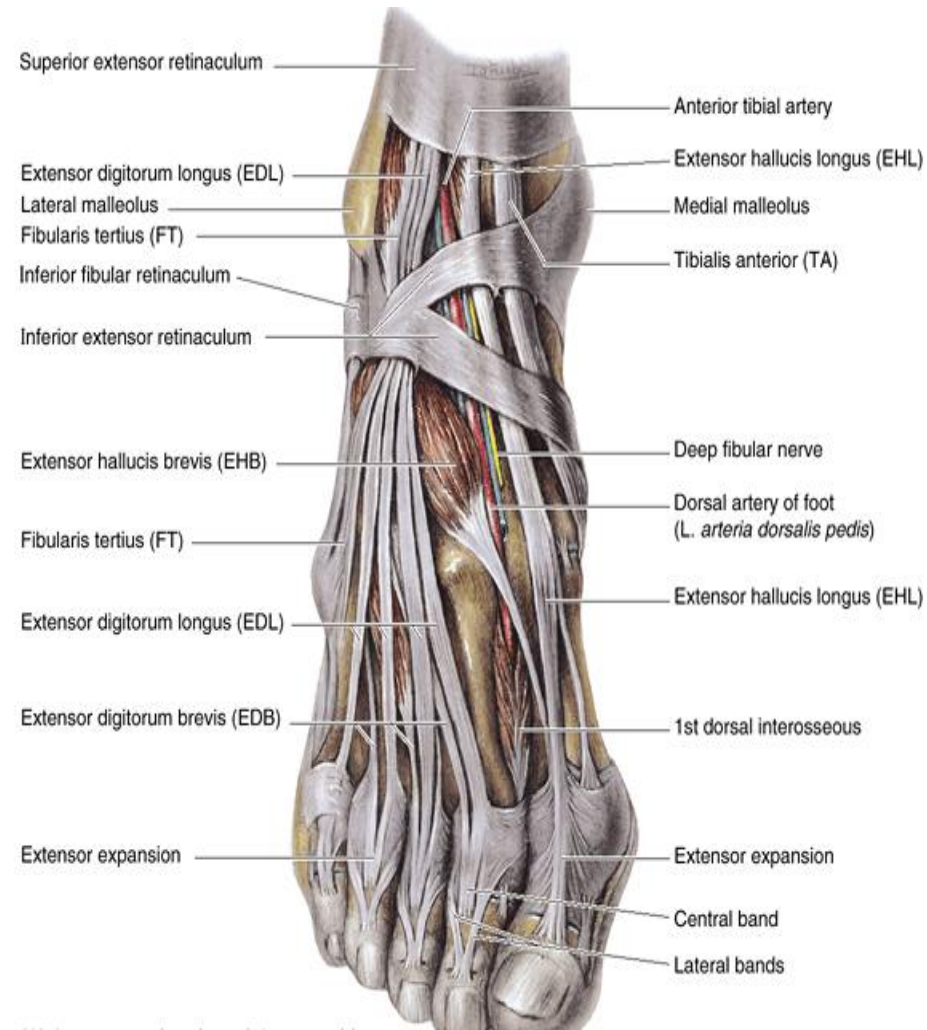
- The deep fascia is thin on the dorsum of the foot, where it is continuous with the inferior extensor retinaculum
- Over the lateral and posterior aspects, it is continuous with the plantar fascia, which has a thick central part (the **plantar aponeurosis**) and weaker medial and lateral parts
- The plantar fascia **holds parts of the foot together**, helps protect the sole from injury, and helps support the longitudinal arches of the foot
- Its intermuscular septum divide the sole in to three compartment
 - Medial
 - Central
 - Lateral compartment of the sole



**(B) Plantar aspect or sole of foot
(inferior view)**

INTRINSIC MUSCLES OF THE FOOT

- ❖ Originate and insert in the foot.
- ❖ Intrinsic muscles mainly **modify the actions of the long** tendons and generate **fine movements** of the toes.
- ❖ There are **two intrinsic muscle** on the dorsum of the foot-extensor digitorum brevis & extensor hallucis brevis.

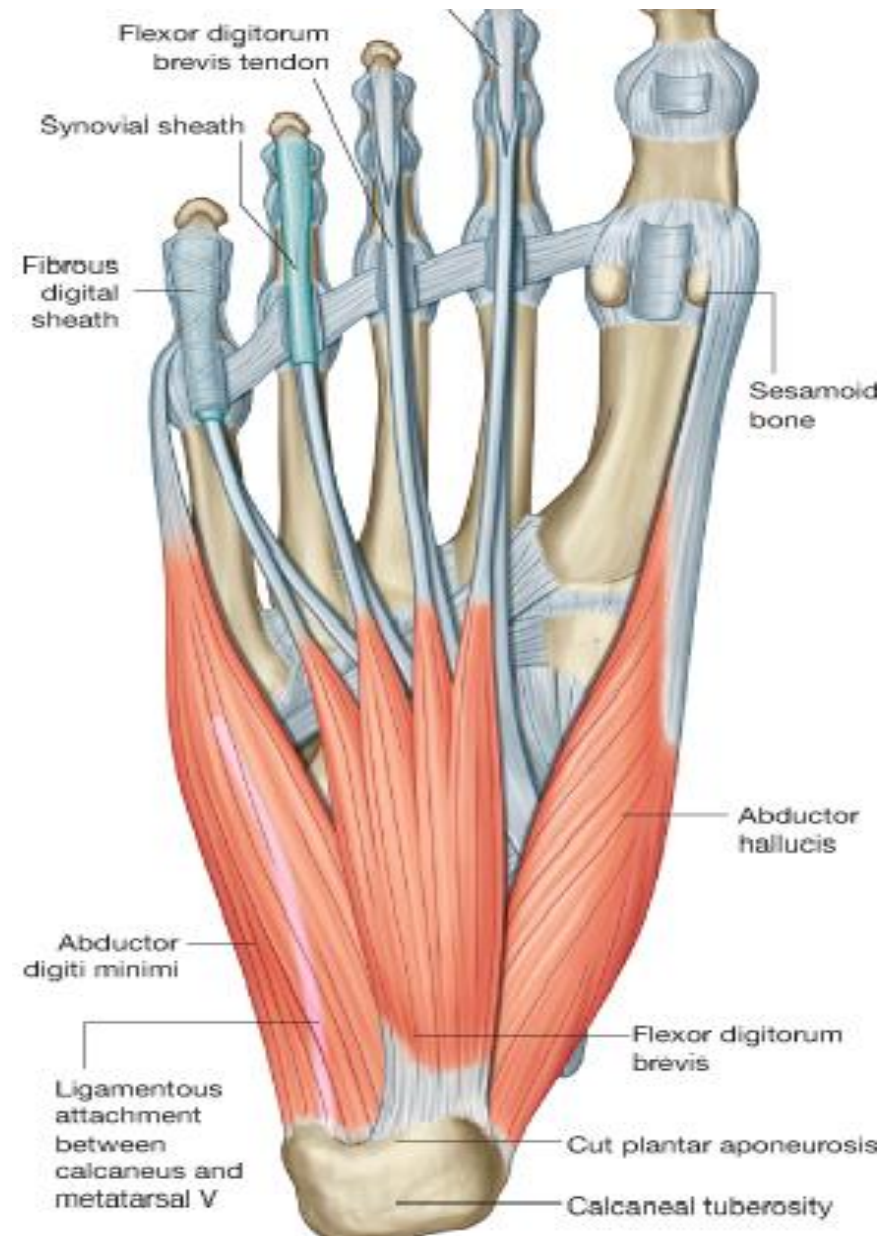


Extensor digitorum brevis-on the dorsal aspect of the foot

Muscle	Proximal attachment	Distal attachment	Innervation	action
Extensor digitorum and hallucis brevis	Superolateral surface of the Calcaneus	Base of proximal phalanx of great toe and tendons of extensor digitorum longus of toes II to IV	Deep fibular nerve	Aid the extensor longus muscles

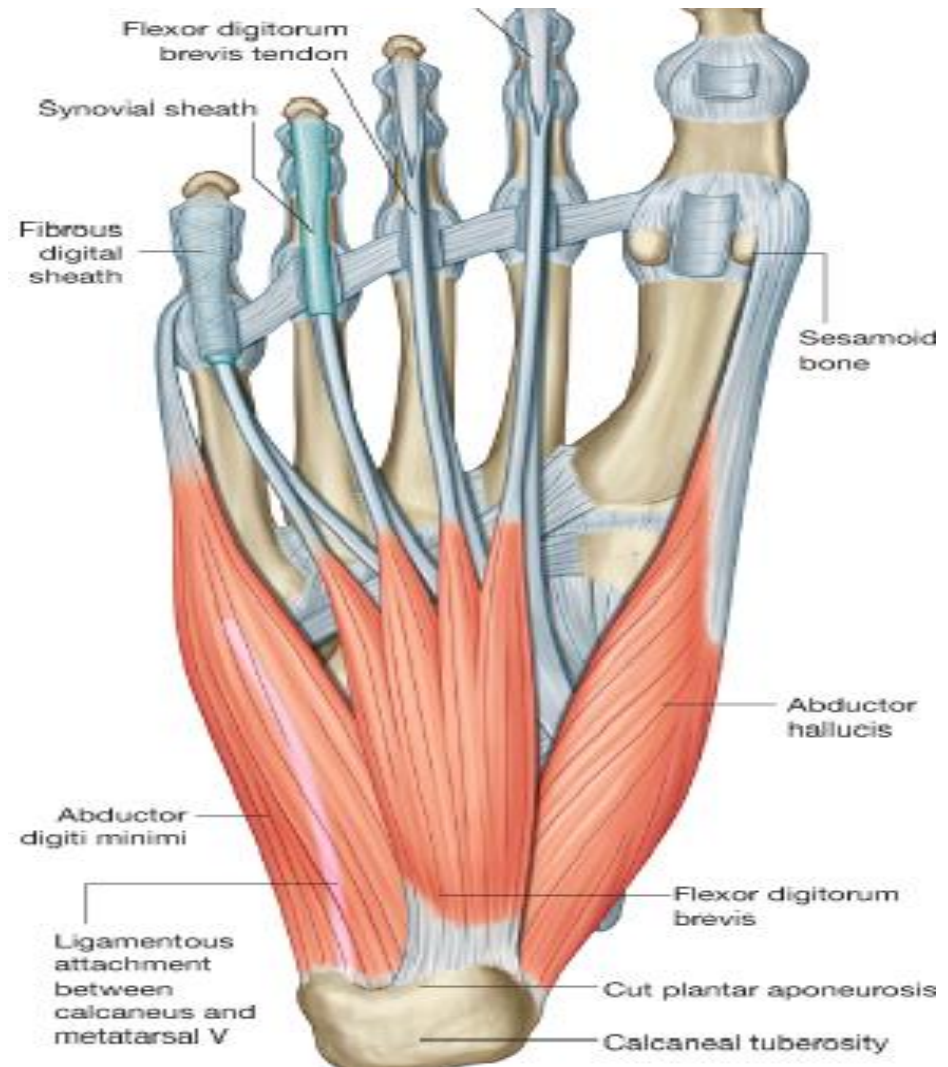
MUSCLES IN THE SOLE

- The muscles in the sole of the foot are organized into four layers
- The plantar muscles maintain the arches of the foot
- They also refine further the efforts of the long muscles, producing **supination** and **pronation** in enabling the foot to adjust to uneven ground



FIRST LAYER

- There are **three** components in the first layer of muscles
- From medial to lateral, these muscles are
 1. abductor hallucis,
 2. flexor digitorum brevis, and
 3. abductor digiti minimi.

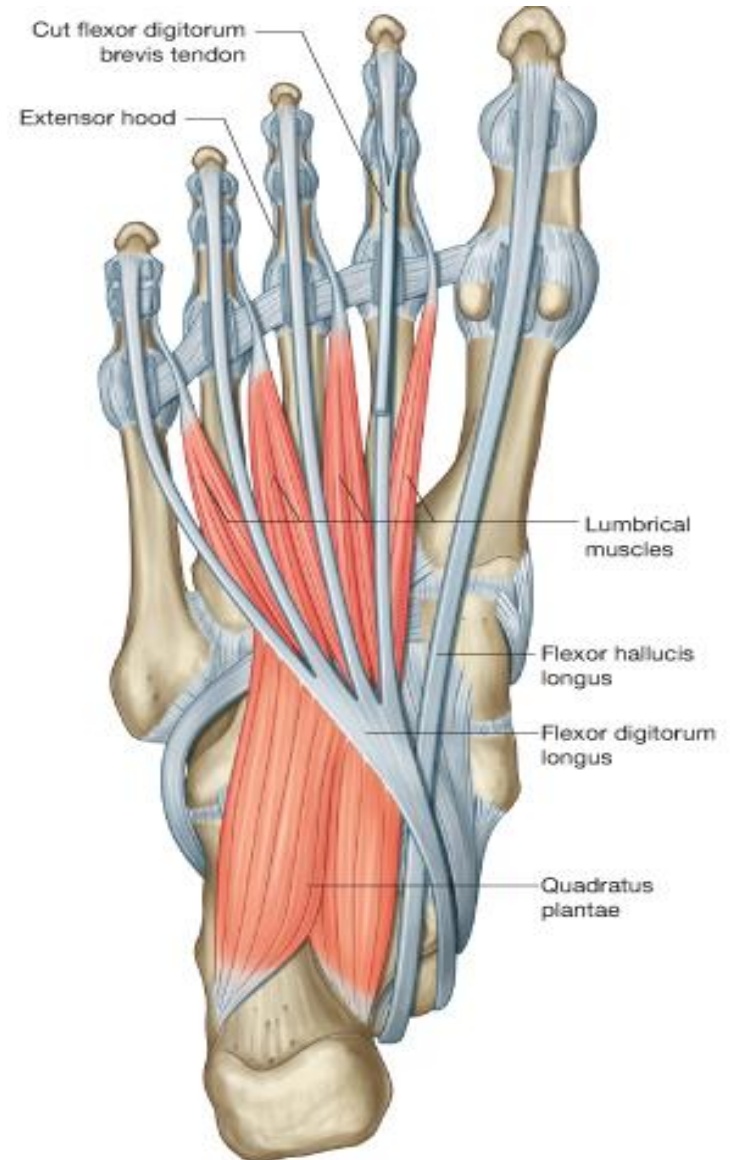


Muscles in the first layer of the sole

Muscle	Proximal attachment	Distal attachment	Innervation	action
Abductor hallucis	Medial process of calcaneus	base of proximal phalanx of great toe	Medial plantar nerve	Abducts and flexes great toe at metatarsophalangeal joint
Flexor digitorum brevis	Medial process of calcaneus and plantar aponeurosis	middle phalanges of lateral four toes	Medial plantar nerve	Flexes lateral four toes at proximal interphalangeal joint
Abductor digiti minimi	Lateral and medial processes of calcaneus and Plantar aponeurosis	base of proximal phalanx of little toe	Lateral plantar nerve	Abducts little toe at the metatarsophalangeal joint

SECOND LAYER

- Consists of the
 1. Quadratus plantae and
 2. Four lumbrical muscles

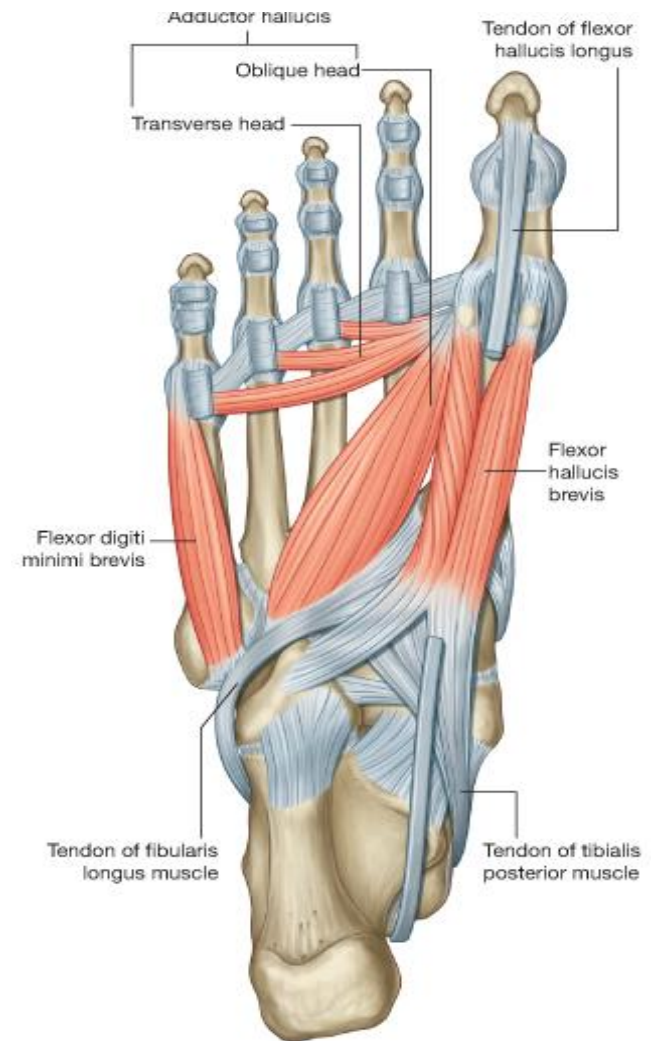


Muscles in the second layer of the sole

Muscles	Proximal attachment	Distal attachment	innervation	action
Quadratus plantae	Medial and lateral surface of calcaneus	tendon of flexor digitorum longus	Lateral plantar nerve	Assists flexor digitorum longus tendon in flexing toes II to V
Lumbricals	tendon of flexor digitorum longus	extensor expansion of toes II to V	First lumbrical- medial plantar nerve 2nd , 3rd and 4th lumbricals- lateral plantar nerve	Flexion of metatarsophalangeal joint and extension of interphalangeal joints

THIRD LAYER

- Three muscles
- ✓ **Flexor hallucis brevis**
 - Arise fro cuboid and lateral cuneiform and insert in to base of proximal phalanx of the great toe
- ✓ **Adductor hallucis**
 - Extend from metatarsals bones 2-4 to base of proximal phalanx of great toe
- ✓ **Flexor digiti minimi brevis**
 - Extend from metatarsals bone 5th to base of proximal phalanx of little toe
- All are innervated by **Lateral plantar nerve except FHB(m.plantar)**
- They are adductor and flexor of the respective digits



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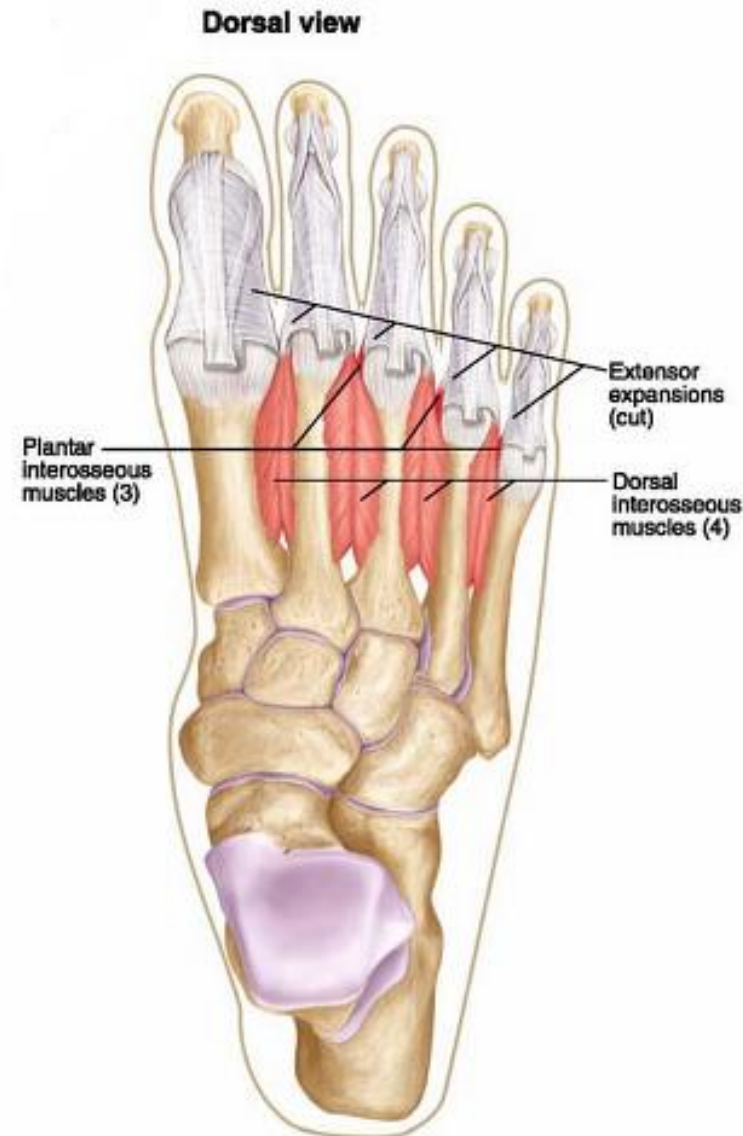
Muscles in the third layer of the sole

Muscle	Proximal attachment	Distal attachment	Innervation	action
Flexor hallucis brevis	Plantar surface of cuboid and lateral cuneiform	Both sides of base of proximal phalanx of the great toe	Medial plantar nerve	Flexes metatarsophalangeal joint of the great toe
Adductor hallucis	<i>oblique head-</i> bases of metatarsals 2-4 <i>Transverse head-</i> ligaments of metatarsophalangeal joints	Lateral side of base of proximal phalanx of great toe	Lateral plantar nerve	Adducts great toe at metatarsophalangeal joint
Flexor digiti minimi brevis	Base of metatarsal V and related sheath of fibularis longus tendon	base of proximal phalanx of little toe	Lateral plantar nerve	Flexes little toe at metatarsophalangeal joint

FOURTH LAYER

❖ The dorsal interossei

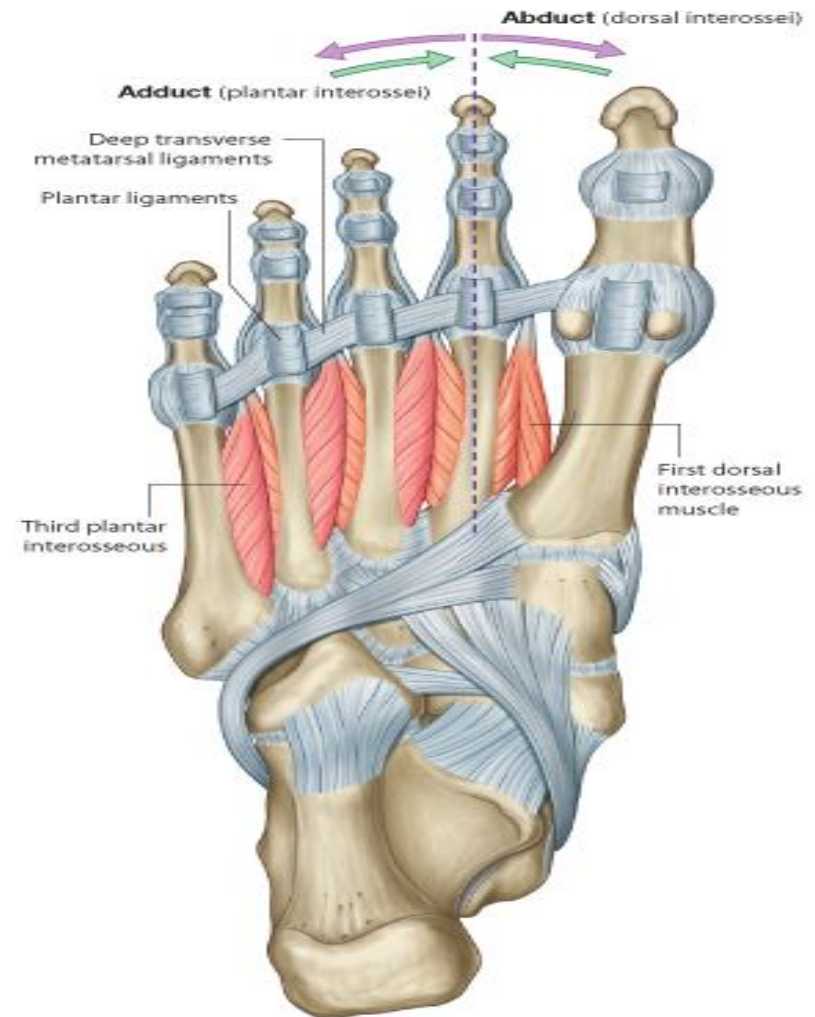
- Extend from Sides of adjacent metatarsals to bases of proximal phalanges of toes II to IV
- Are **abductor** and flexor of 2-4 toes
- ❑ Are innervated by **Lateral plantar nerve**



FOURTH LAYER cont...

Plantar interossei-3

- Extend from medial Sides of adjacent 3-5 metatarsals to bases of proximal phalanges of 3-5 toes
- Are **adductor** and flexor of 3-5 toes
- Are innervated by **Lateral plantar nerve**



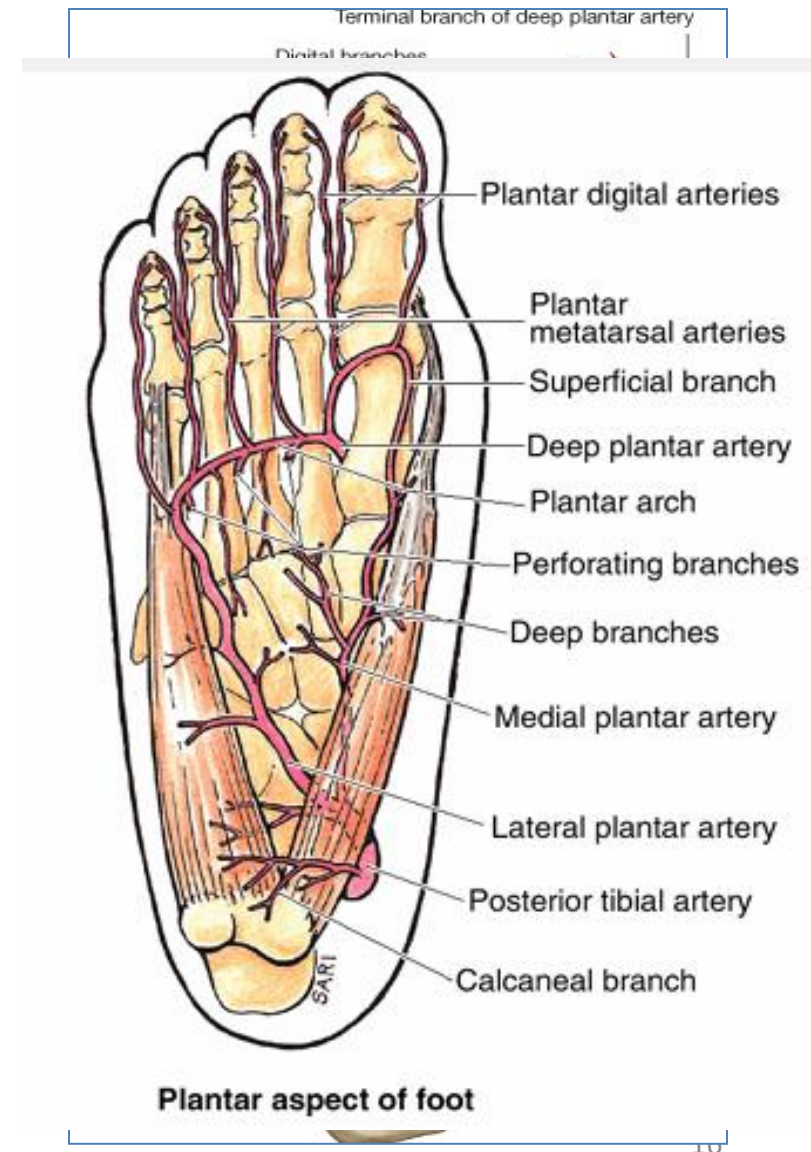
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Muscles in the fourth layer of the sole

Muscles	Proximal attachment	Distal attachment	innervation	action
Dorsal interossei	Sides of adjacent metatarsals	Dorsal expansions and bases of proximal phalanges of toes II to IV	Lateral plantar nerve	Abduction of toes II to IV Flex metatarsophalangeal joint
Plantar interossei	Medial sides of metatarsals of toes III to V	Dorsal expansions and bases of proximal phalanges of toes III to V	Lateral plantar nerve	Adduction of toes III to V Flex the metatarsophalangeal joint

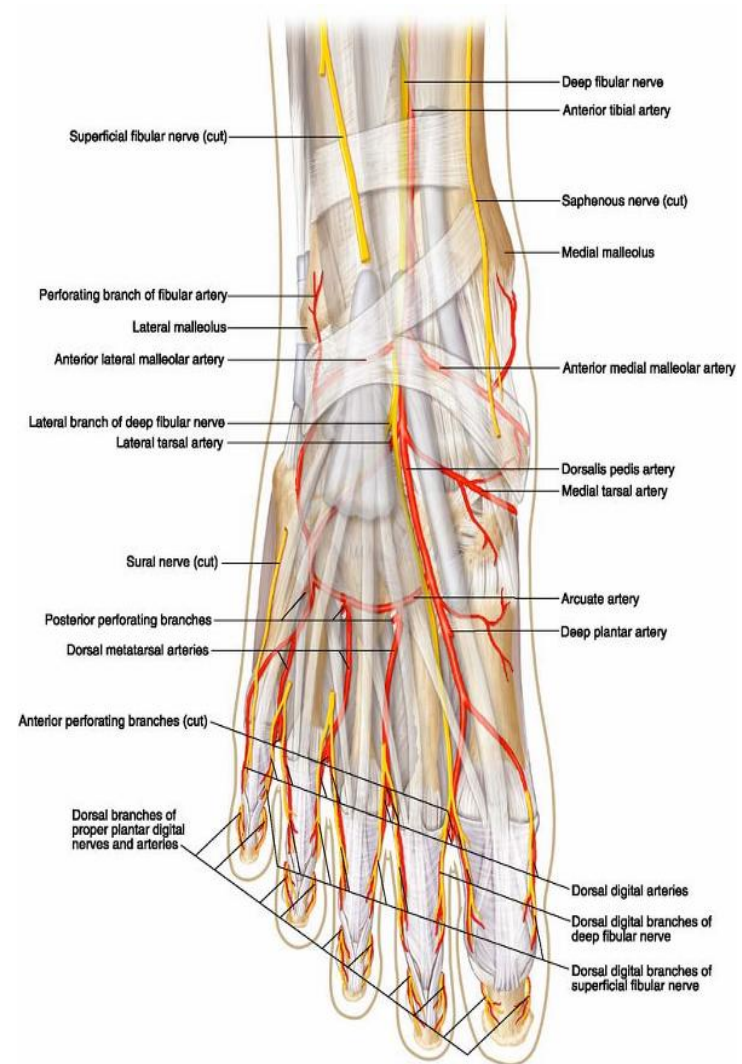
ARTERIES OF THE FOOT

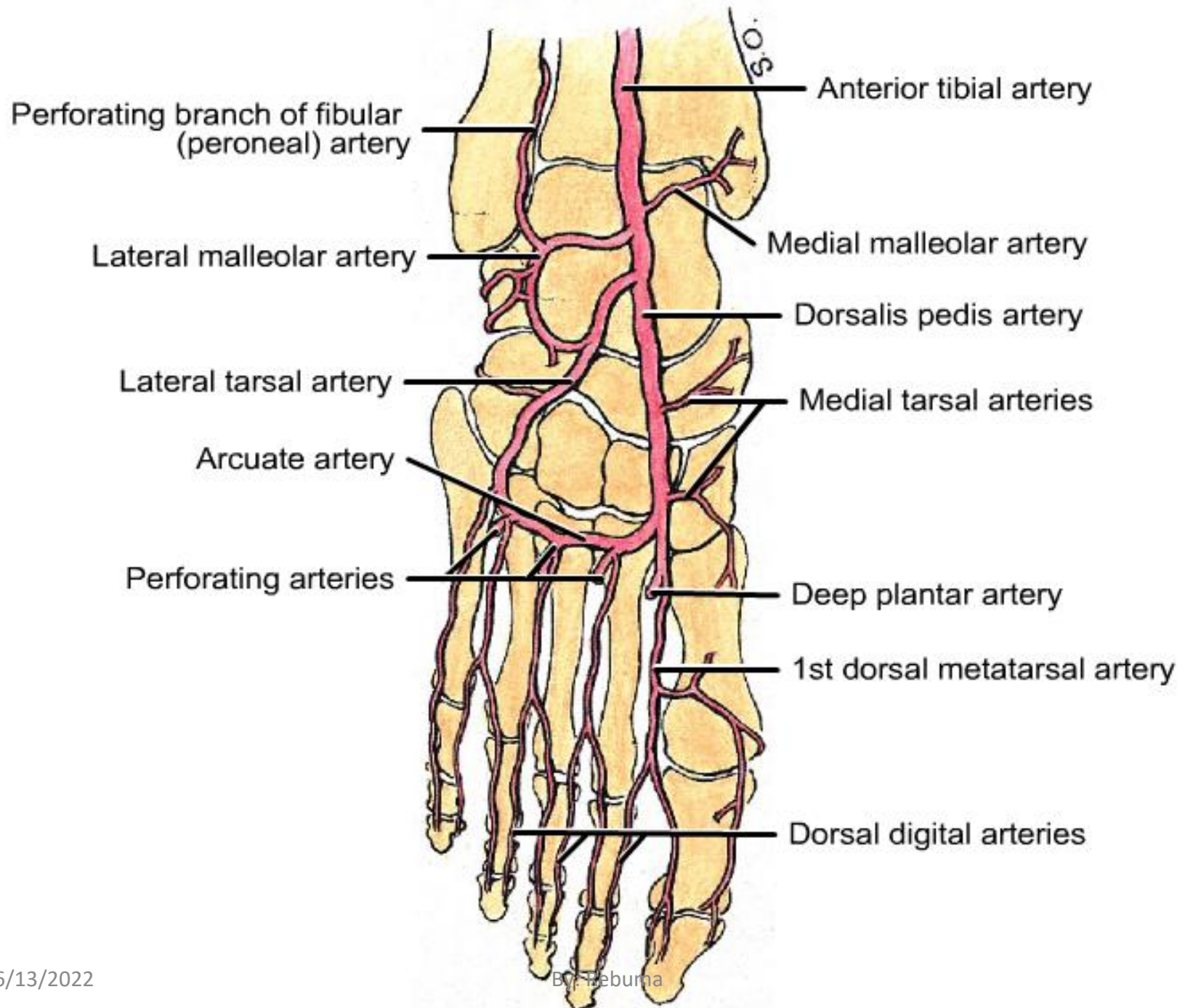
- Blood supply to the foot is by branches of the **posterior tibial** and **dorsalis pedis** (dorsal artery of the foot) arteries.
- The **posterior tibial artery** enters the sole and bifurcates into **lateral** and **medial** plantar arteries.
- The **lateral plantar** artery joins with the terminal end of the **dorsalis pedis artery** (the deep plantar artery) to form the **deep plantar arch**.



ARTERIES OF THE FOOT

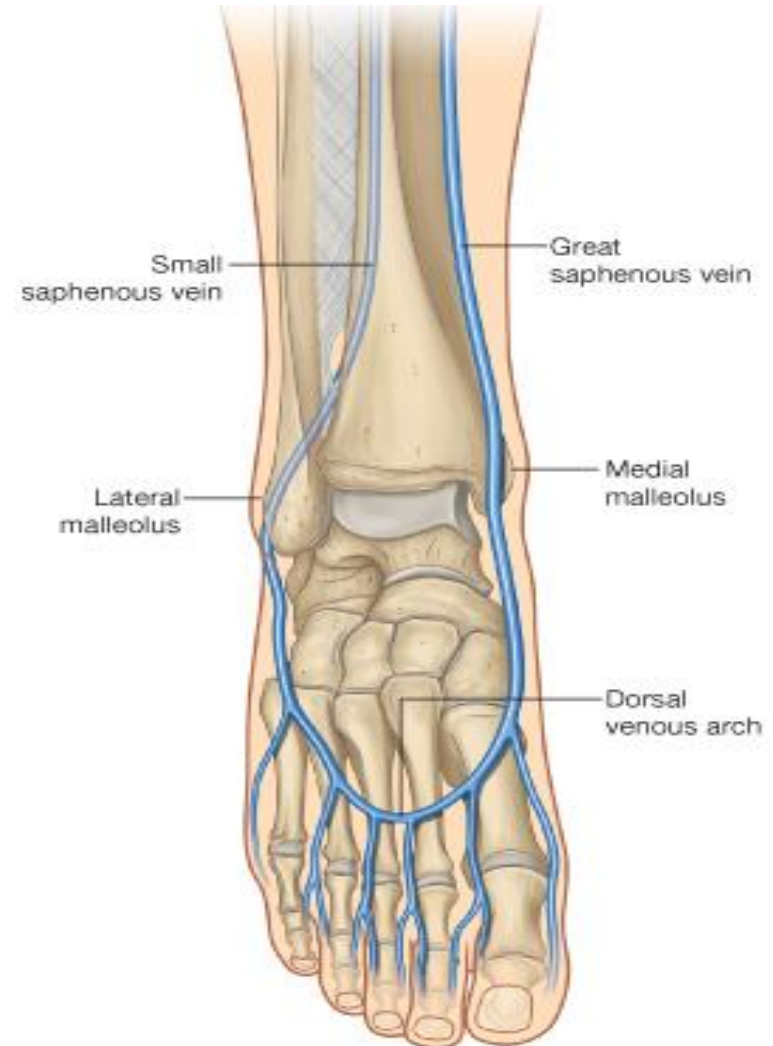
- The **dorsalis pedis** artery is the continuation of the **anterior tibial artery**, passes on the dorsal aspect of the foot and then inferiorly, as the **deep plantar artery**, between metatarsals I and II to enter the sole of the foot.
- Branches of the dorsalis pedis artery include **lateral** and **medial tarsal branches**, an **arcuate artery**, and a first **dorsal metatarsal** artery.





VEINS OF THE FOOT

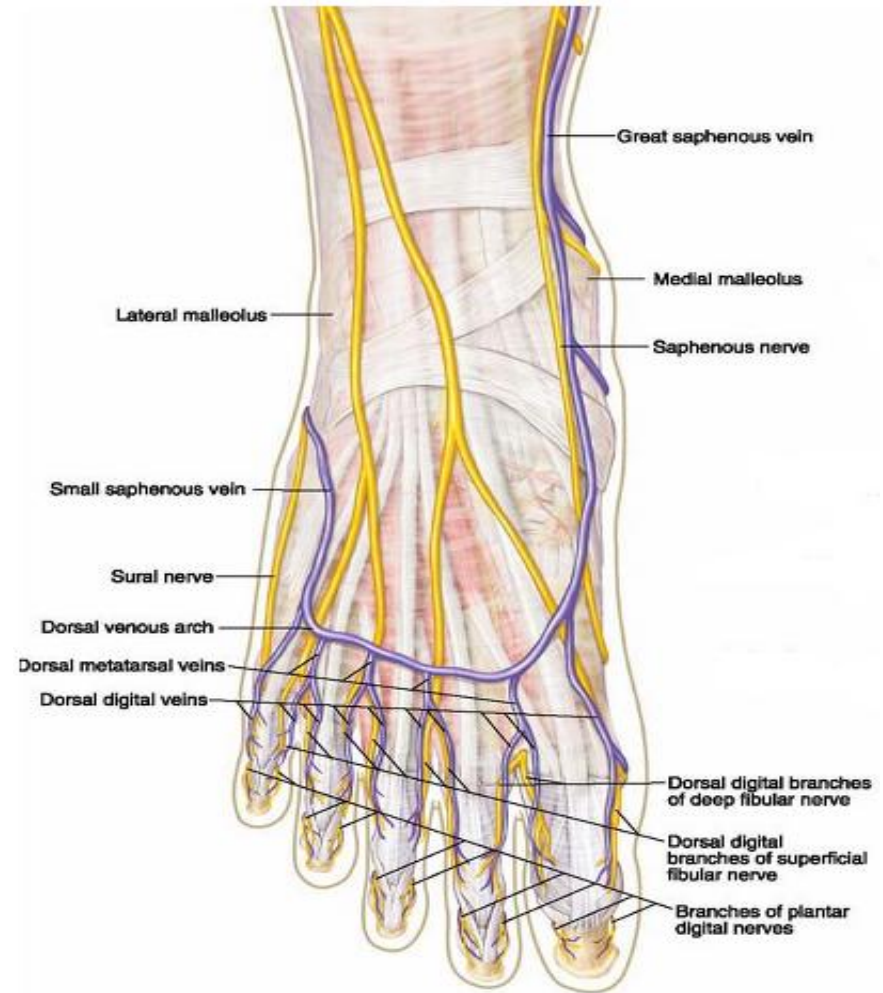
- There are interconnected networks of **deep** and **superficial veins** in the foot.
- The deep veins follow the arteries.
- Superficial veins drain into a **dorsal venous arch**
- The **great saphenous** vein originates from the medial side of the arch and passes anterior to the medial malleolus and onto the medial side of the leg;
- The **small saphenous** vein originates from the lateral side of the arch and passes posterior to the lateral malleolus and onto the back of the leg.



NERVES OF THE FOOT

The foot is supplied by the

1. Tibial
2. Deep fibular
3. Superficial fibular
4. Sural (origin from tibial and common peroneal)and
5. Saphenous nerves(origin from femoral)

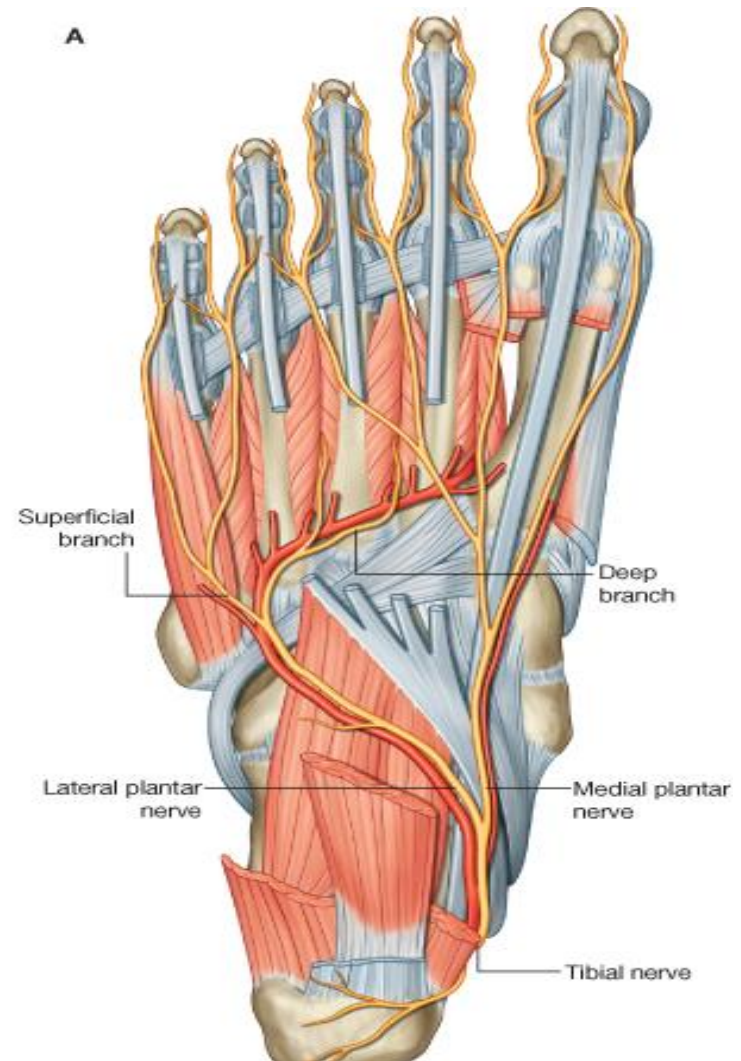


NERVES OF THE FOOT

❖ Tibial nerve

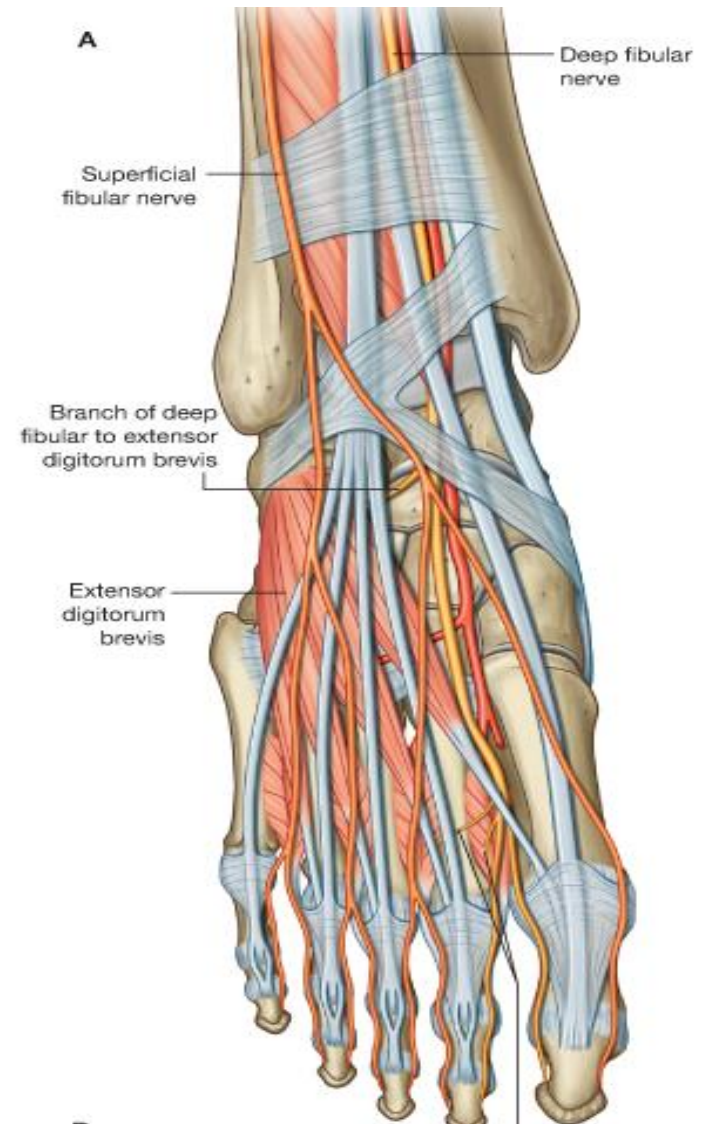
Midway between the medial malleolus and the heel, the tibial nerve bifurcates with the posterior tibial artery into:

- **large medial plantar nerve;**
- **a smaller lateral plantar nerve**

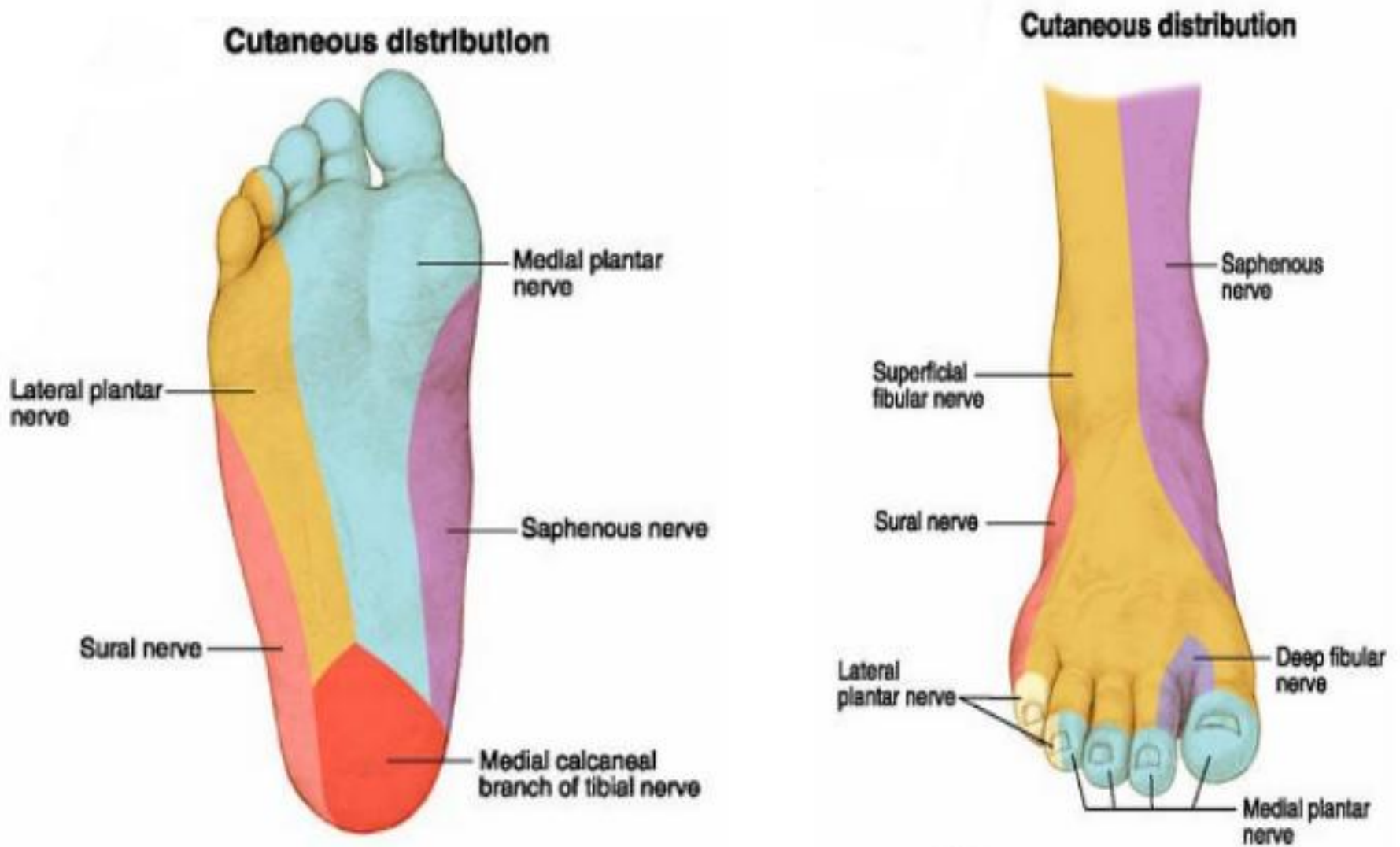


NERVES OF THE FOOT

- The **deep fibular nerve** innervates :
 - ✓ extensor digitorum brevis & extensor hallucis brevis, and
 - ✓ supplies general sensory branches to the skin on the adjacent dorsal sides of the **first and second toes** and to the web space between them.

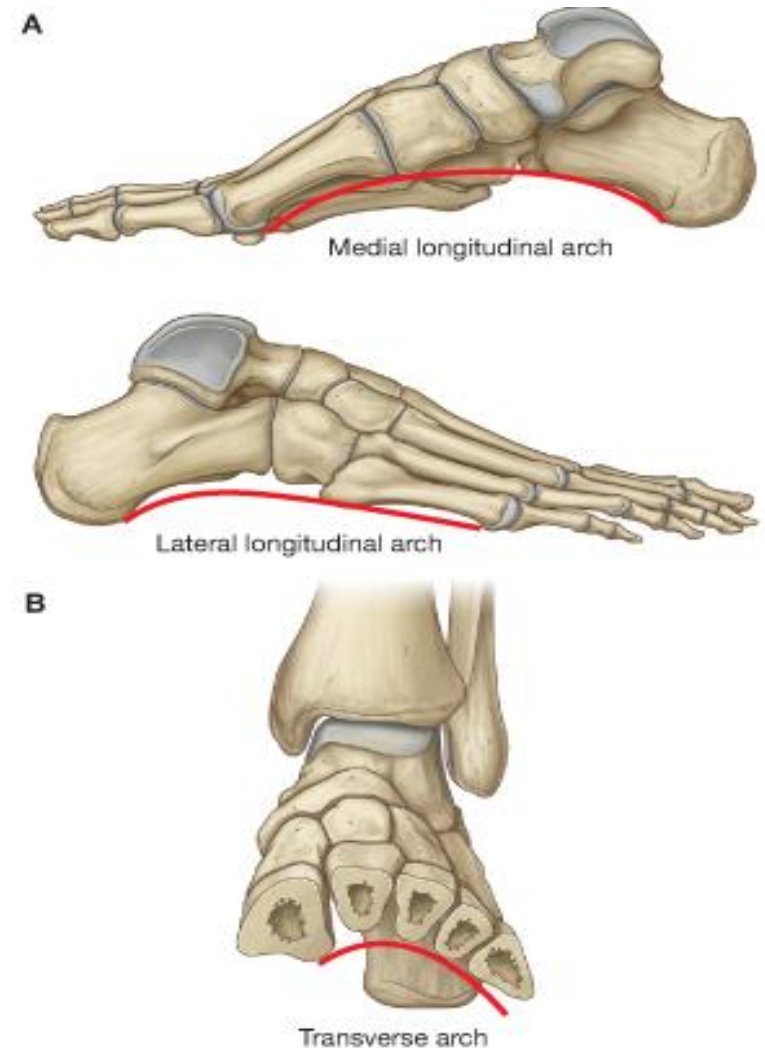


Cutaneous innervation of the foot



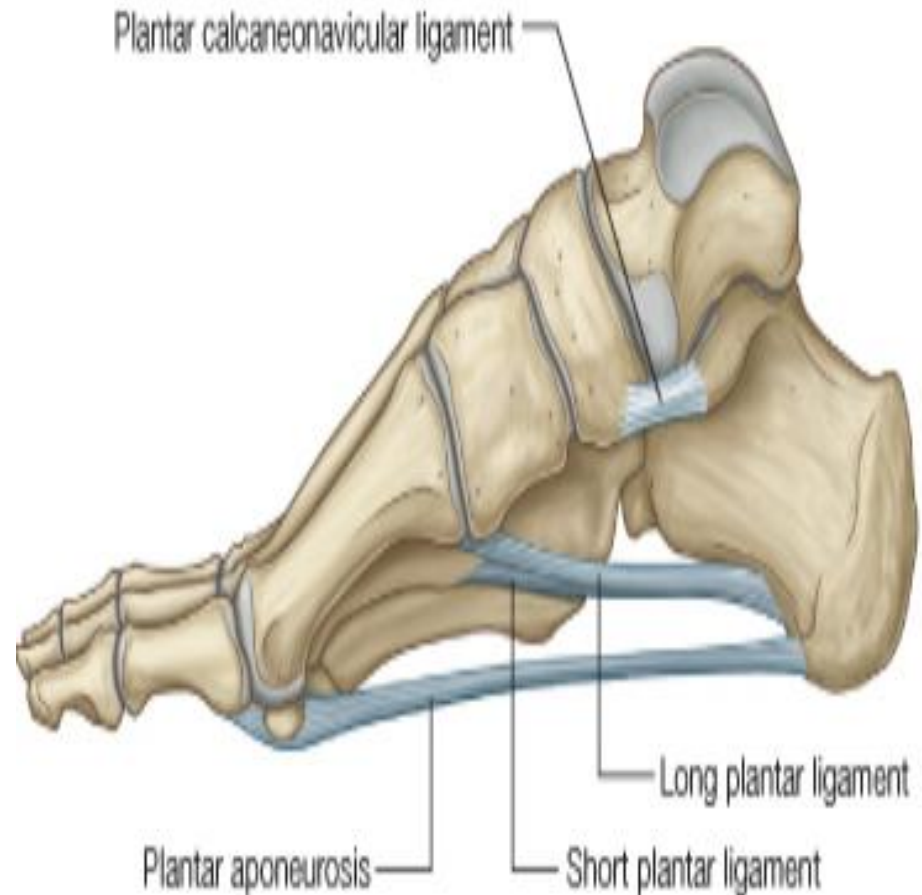
ARCHES OF THE FOOT

- The bones of the foot do not lie in a horizontal plane.
- There are longitudinal and transverse arches which **absorb** and **distribute** downward forces from the body during standing and moving on different surfaces



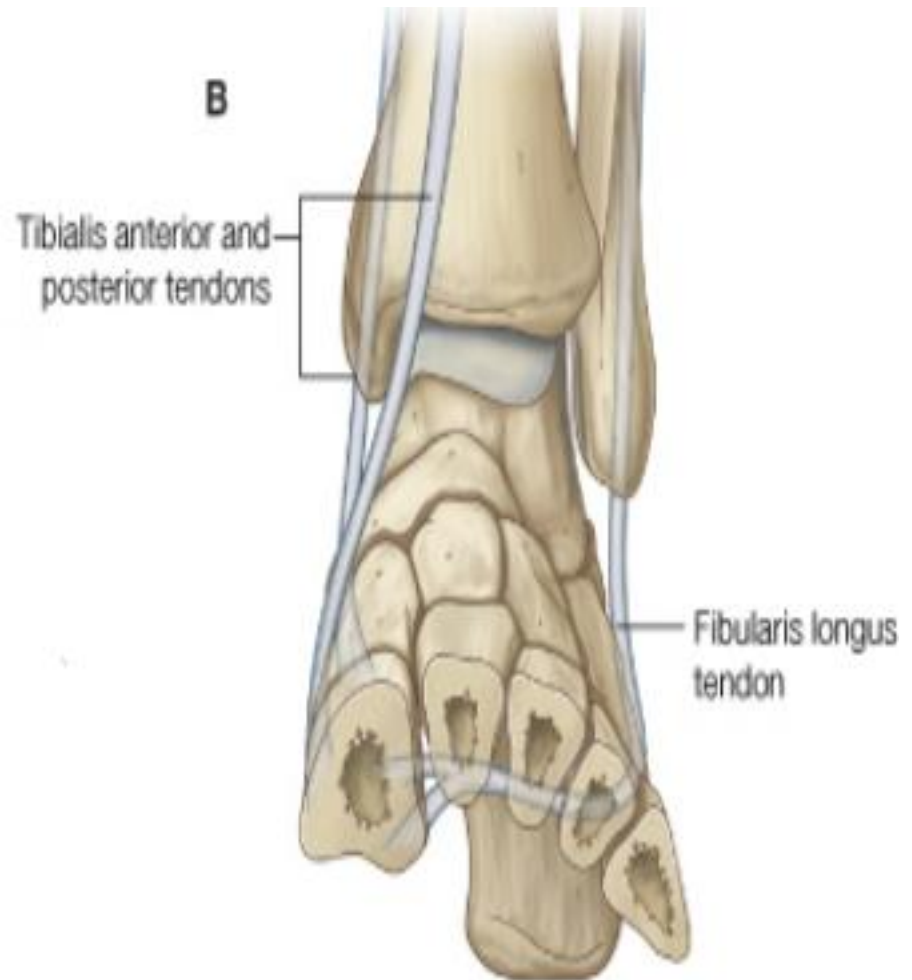
THE LONGITUDINAL ARCH

- The longitudinal arch is formed between the posterior end of the **calcaneus** and the heads of the **metatarsals**
- It is **highest** on the **medial side** and **lowest** on the **lateral side**



THE TRANSVERSE ARCH

- is **highest** in a **coronal plane** that cuts through the head of the talus
- **Disappears** near the **heads of the metatarsals** where these bones are held together by the deep transverse metatarsal ligaments



SUPPORTS OF THE ARCH

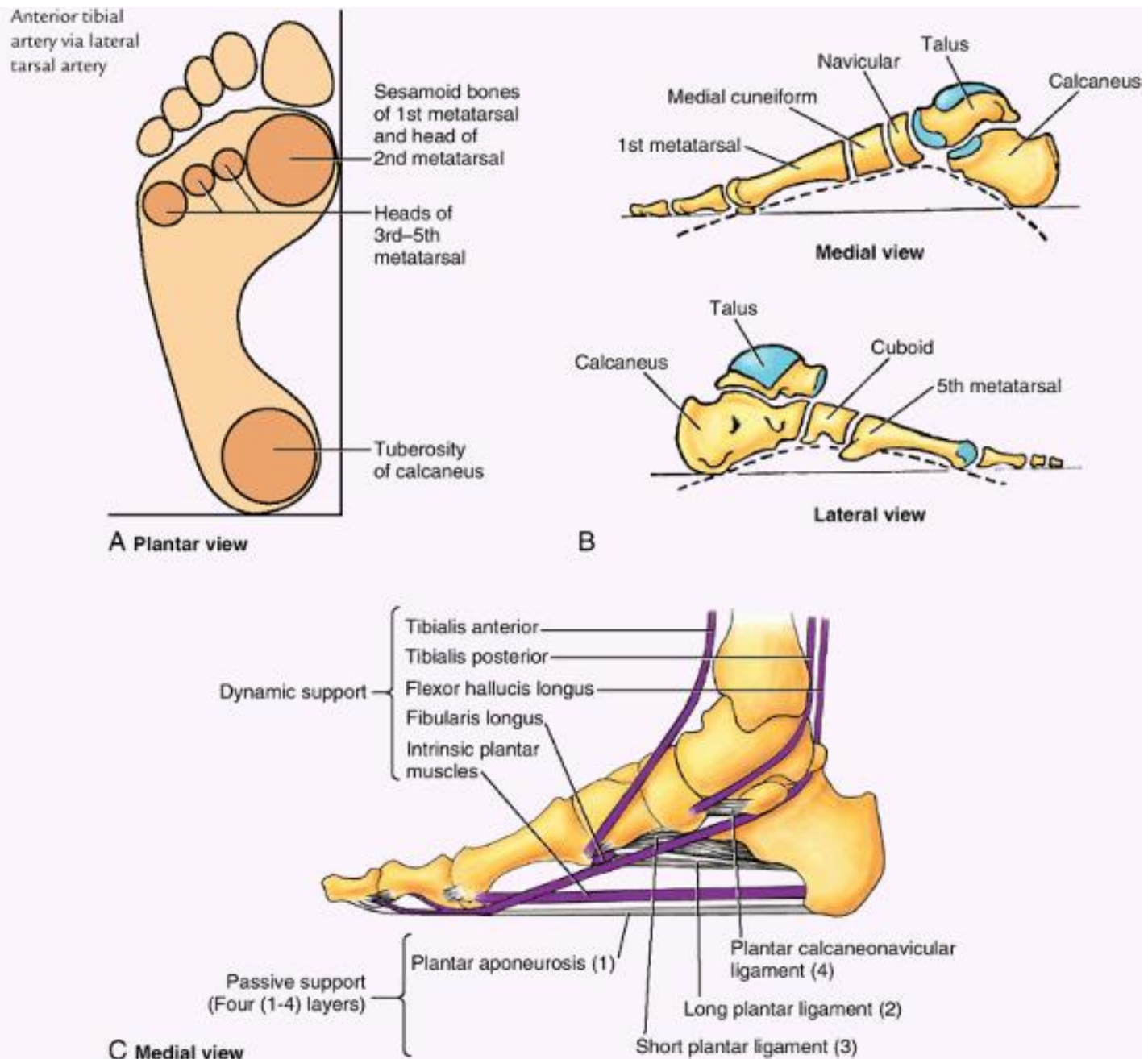
Ligaments and muscles support the arches of the foot

➤ ligaments that support the arches include

1. plantar calcaneonavicular
2. plantar calcaneocuboid
3. long and short plantar ligaments
4. the plantar aponeurosis

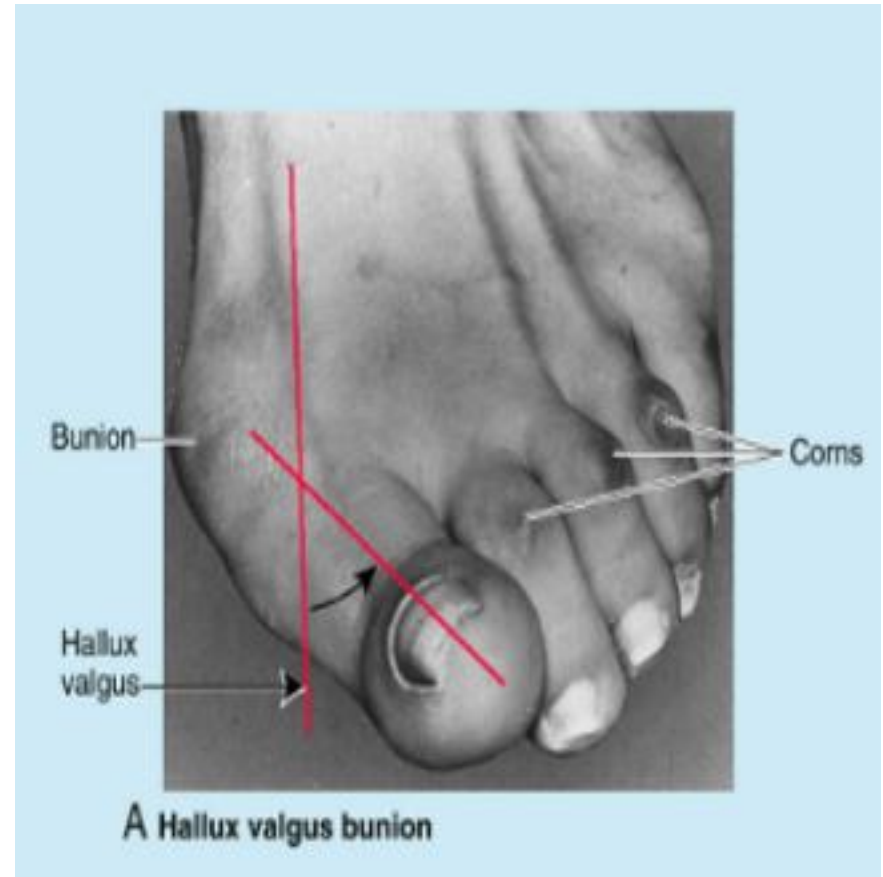
➤ Muscles that provide dynamic support for the arches during walking include

1. the tibialis anterior and posterior, and
2. the fibularis longus, flexor hallucis longus



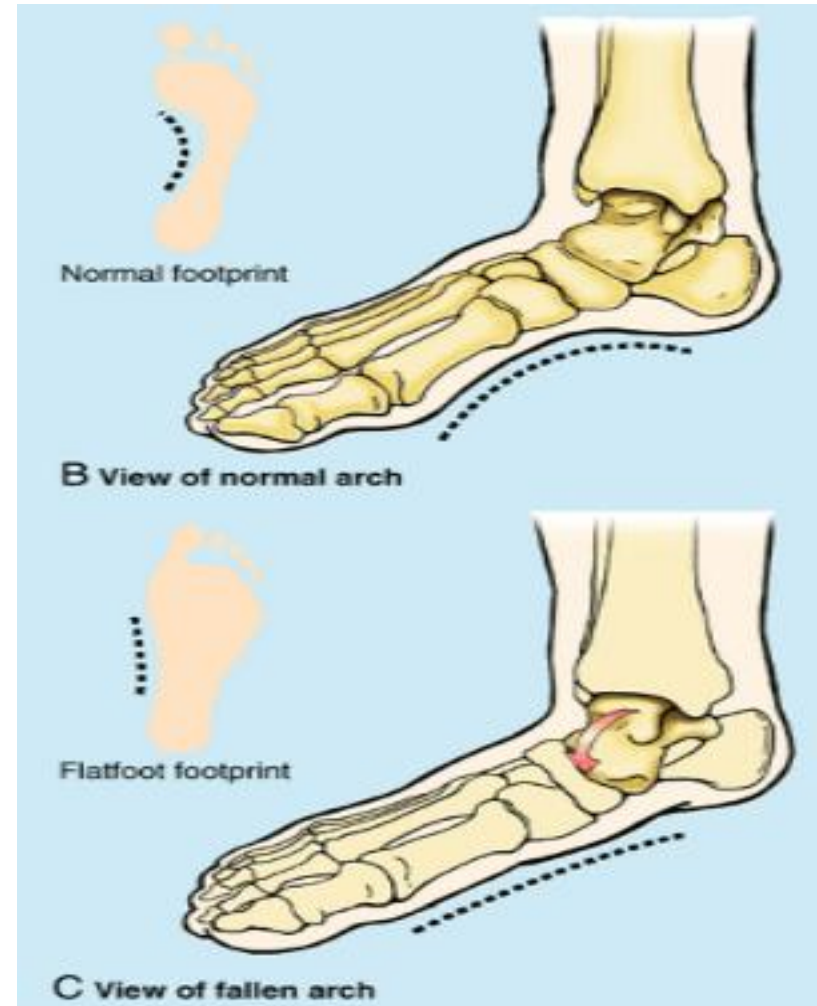
Hallux Valgus

- is a foot deformity caused by pressure from footwear and degenerative joint disease
- It is characterized by lateral deviation of the great toe.



Pes Planus (Flatfeet)

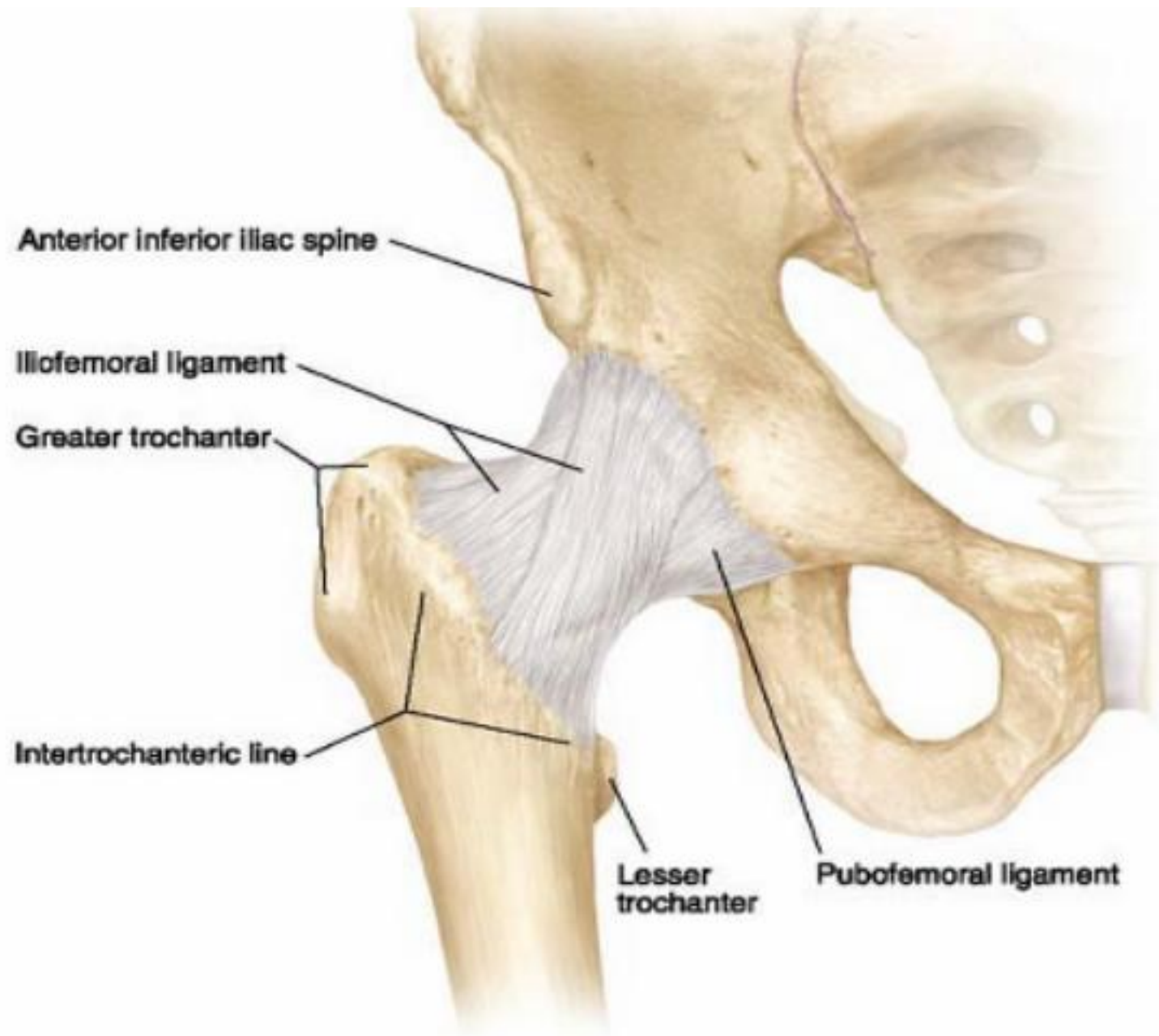
- Acquired flatfeet are likely to be secondary to dysfunction of the tibialis posterior owing to trauma, degeneration with age, or denervation



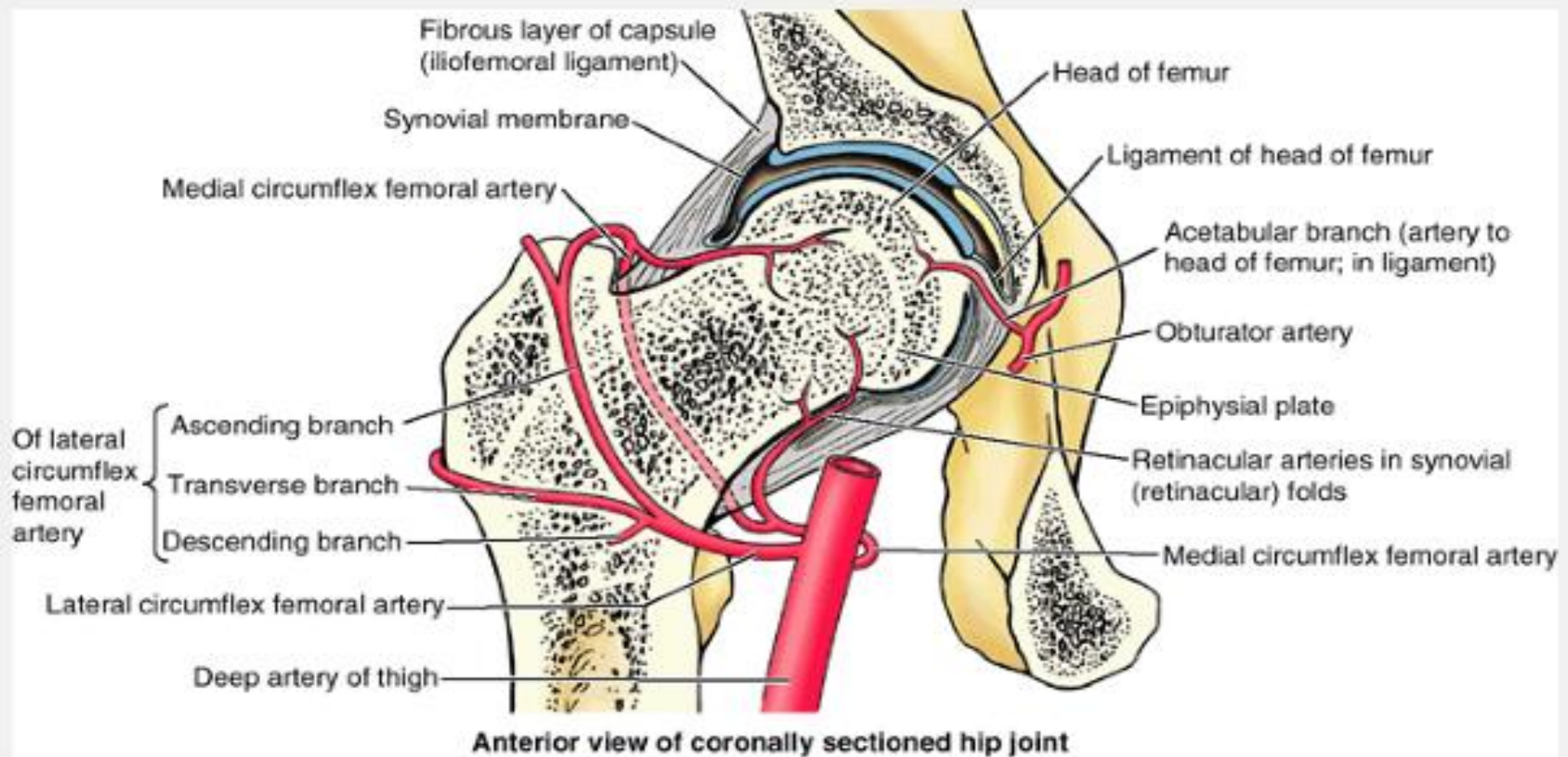
JOINTS OF THE LOWER LIMB

HIP JOINT

- It is a strong, largest and stable multiaxial ball and socket type of synovial joint
- ❖ **Articular Surfaces**
 - The head of the femur articulates with the cup-like acetabulum of the hip bone
 - Joint capsule
 - attached proximally to the margins of the acetabulum and to the transverse acetabular ligament
 - Distally, it is attached to the femoral neck
- ❖ The hip joint is reinforced
 - ✓ Anteriorly and superiorly by the strong Y-shaped iliofemoral ligament :- prevents hyperextension of the hip joint during standing
 - ✓ Inferiorly and anteriorly by the pubofemoral ligament
 - ✓ Posteriorly by the weak ischiofemoral ligament

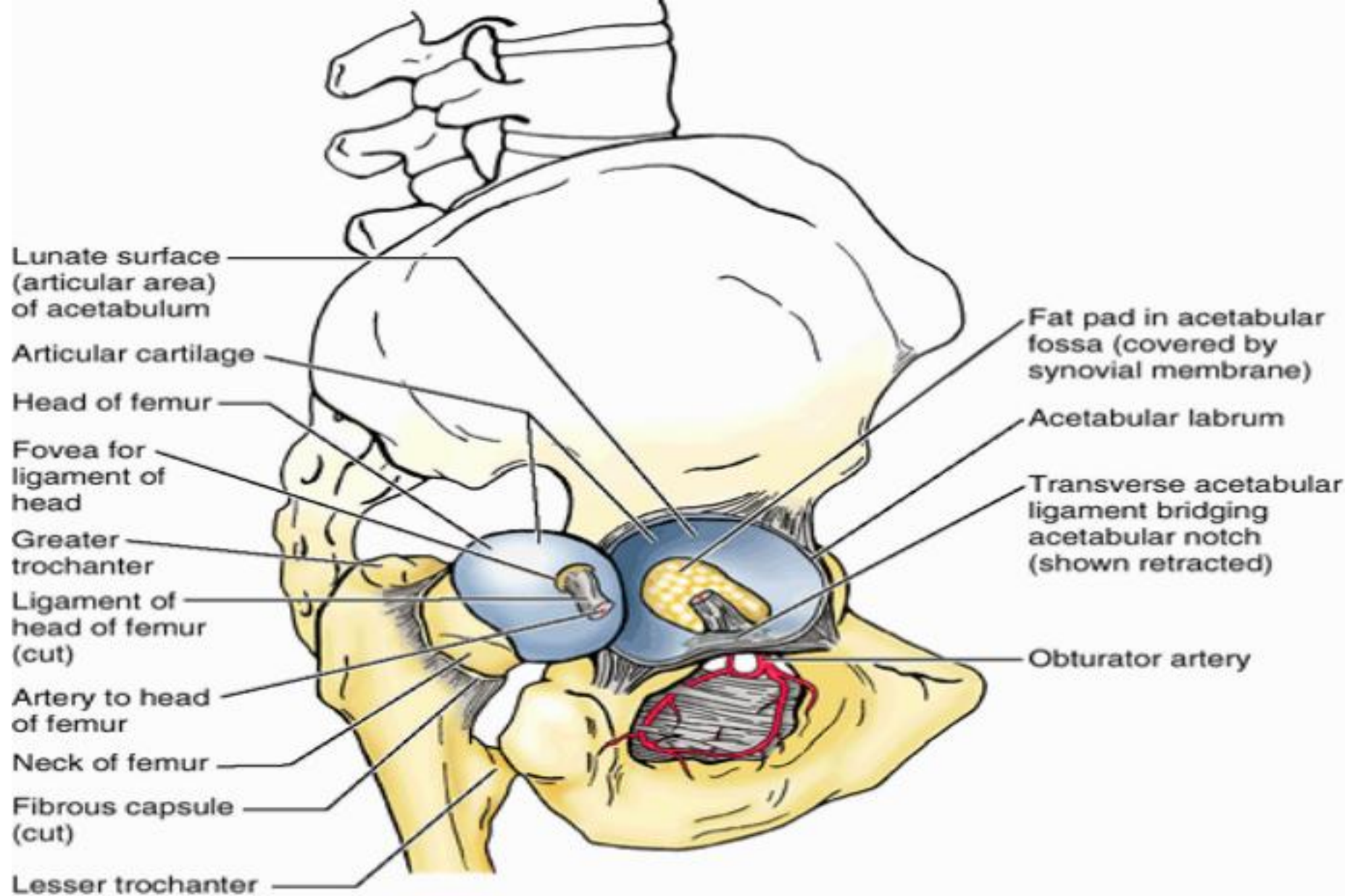


- Blood supply of the hip joint
 - Medial and Lateral femoral circumflex artery
 - Obturator artery
 - Inferior gluteal artery



HIP JOINT...

- Nerve supply of the hip joint
 - Femoral nerve (by the branch to rectus femoris)
 - Obturator nerve
 - Nerve to quadratus femoris
 - Superior gluteal nerve
- Movements of the hip joint
 - Flexion
 - Extension
 - Abduction
 - Adduction
 - Medial and lateral rotation
 - Circumduction



Lateral view

6/13/2022

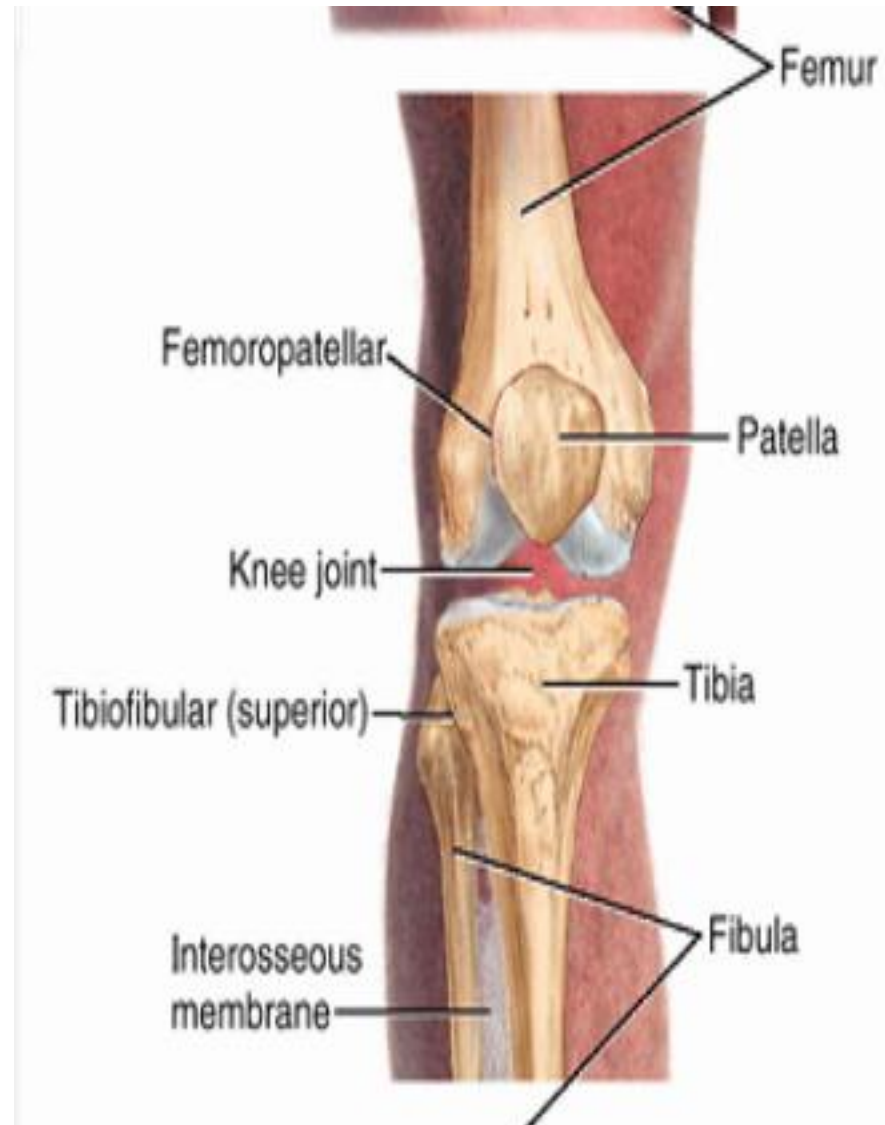
By: Rebuma

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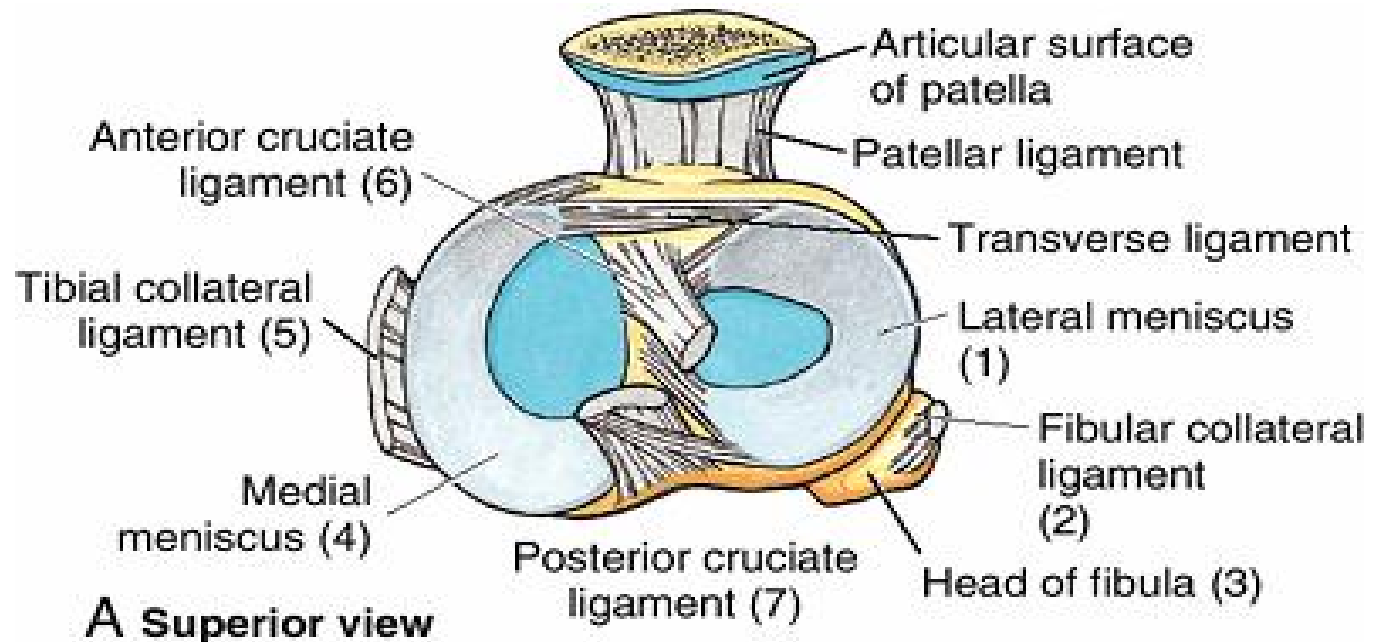
JOINTS CONT...

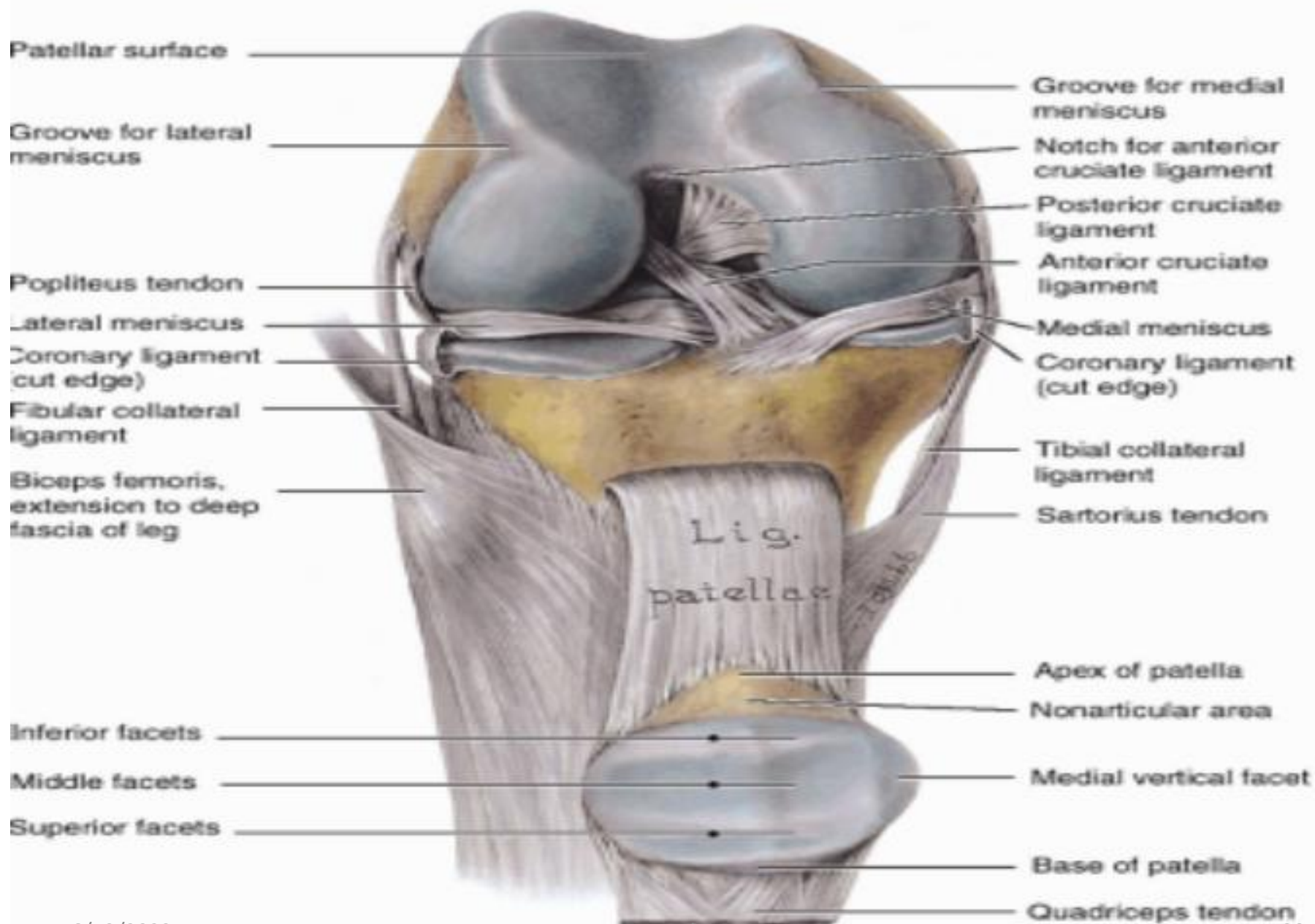
2.KNEE JOINT

- Hinge type of synovial joint
- Movement:- flexion and extension
- The articular surfaces
three articulations:
 - **Two femorotibial** articulations, between the lateral and the medial femoral and tibial condyles
 - **One intermediate femoropatellar** articulation between the patella and the femur



- **Ligaments of the knee**
 - Tibial (medial) collateral ligament
 - Fibular (lateral) collateral ligament
 - Anterior cruciate ligament
 - Posterior cruciate ligament
 - Patellar ligament





(B) Anterior view with knee flexed

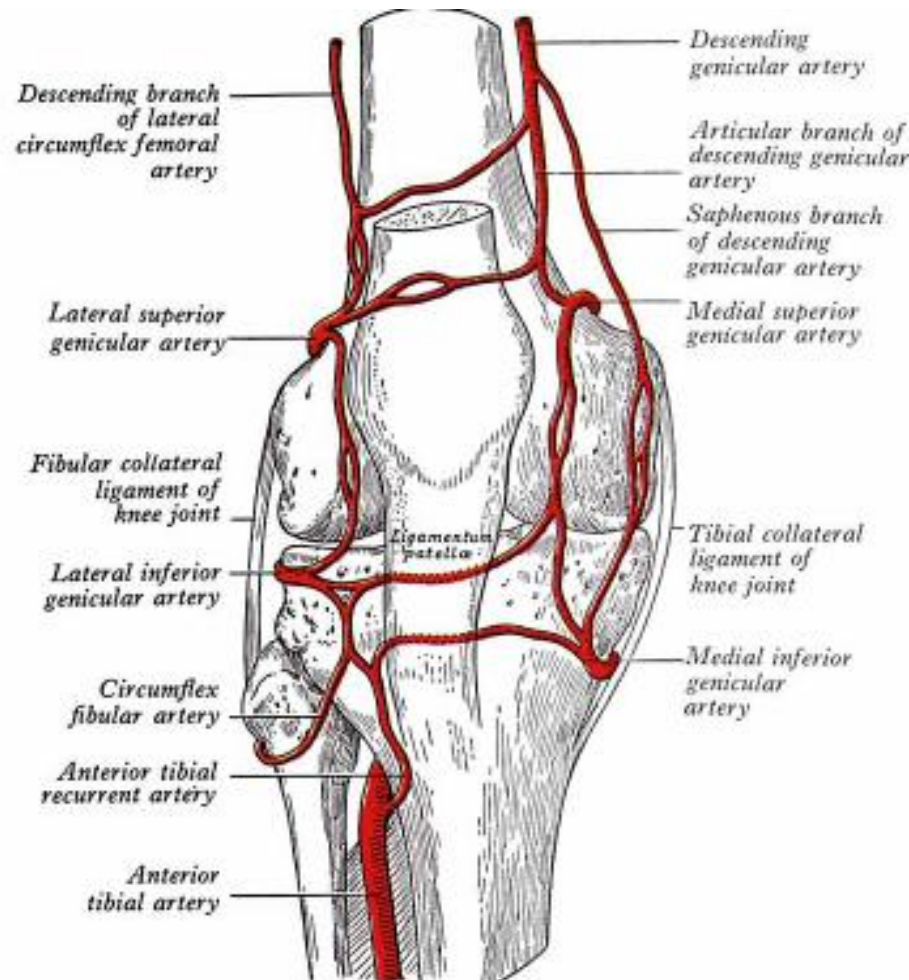
By: Rebuma

- **Blood supply of the knee joint**

- Vessels form genicular anastomosis around the knee
- Genicular branches of
 - Femoral artery
 - Anterior tibial recurrent ar
 - Circumflex fibular artery

- **The nerve supply**

- Articular branches from
 - Femoral
 - Tibial
 - Common fibular

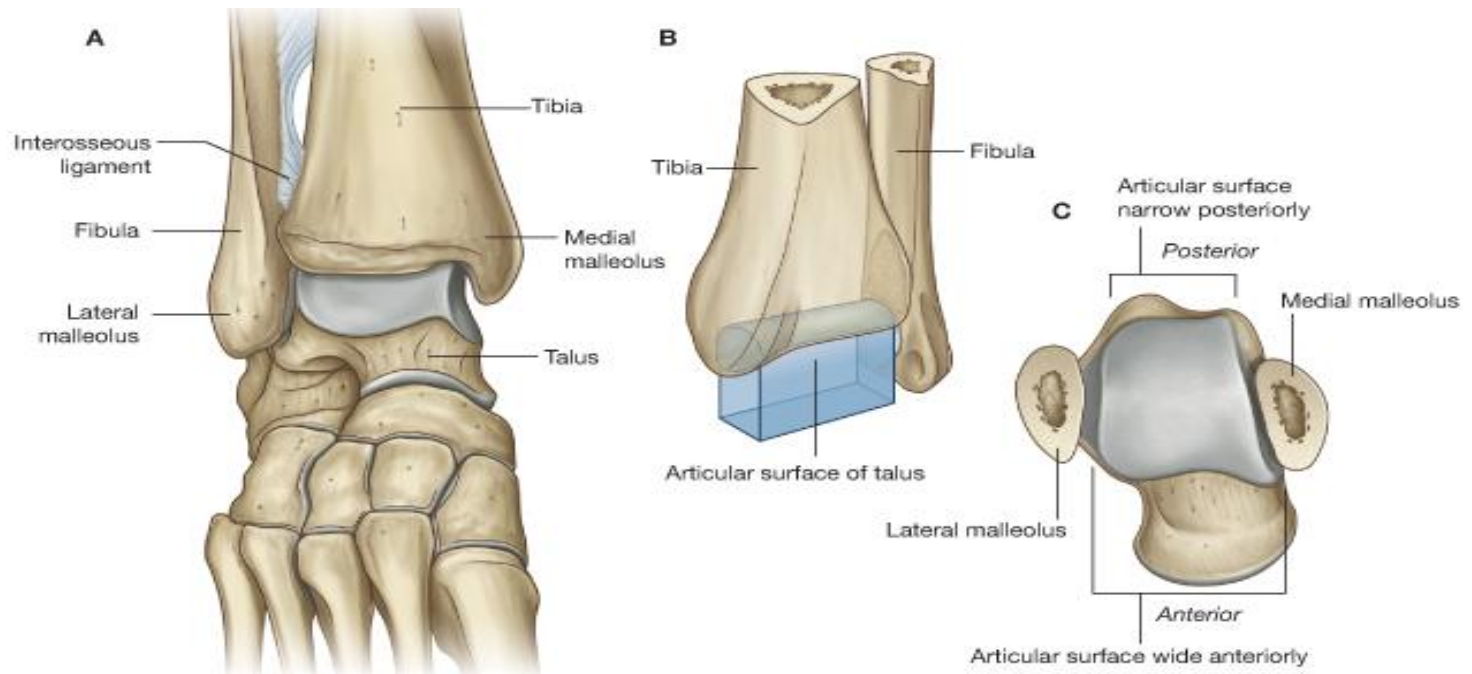


JOINTS CONT...

3 THE ANKLE JOINT

➤ Hinge synovial joint

- Articulation:- **talus** with distal end of **tibia** and **fibula**
- Movements :- **dorsiflexion** and **plantarflexion** of the foot

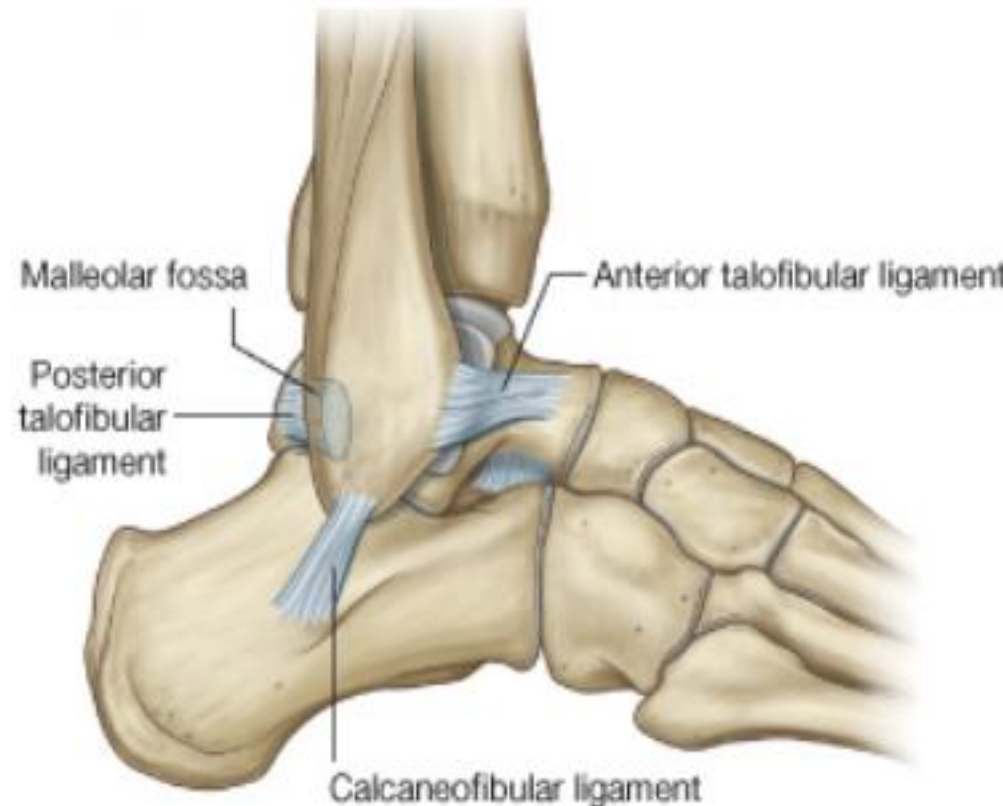


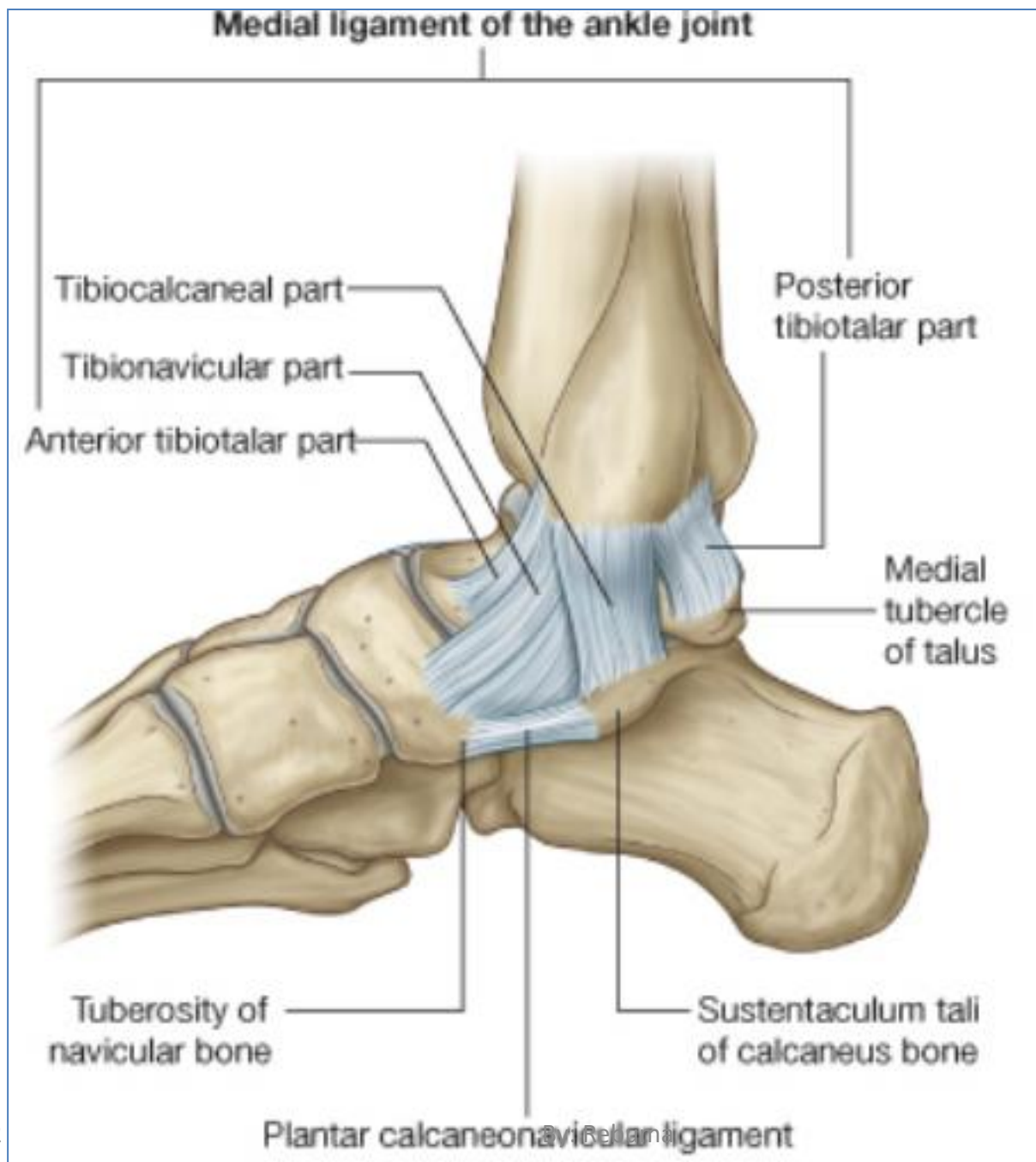
Ligaments of the ankle

- Lateral ligament
 - Anterior talofibular ligament
 - Posterior talofibular ligament
 - Calcaneofibular ligament
- Medial ligament
- **Blood supply**
 - Anterior and posterior tibial arteries

Nerve supply

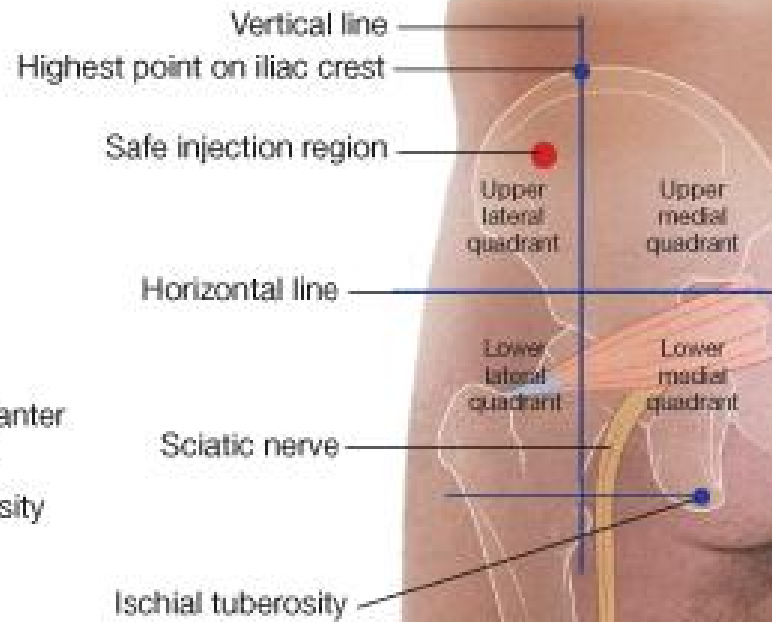
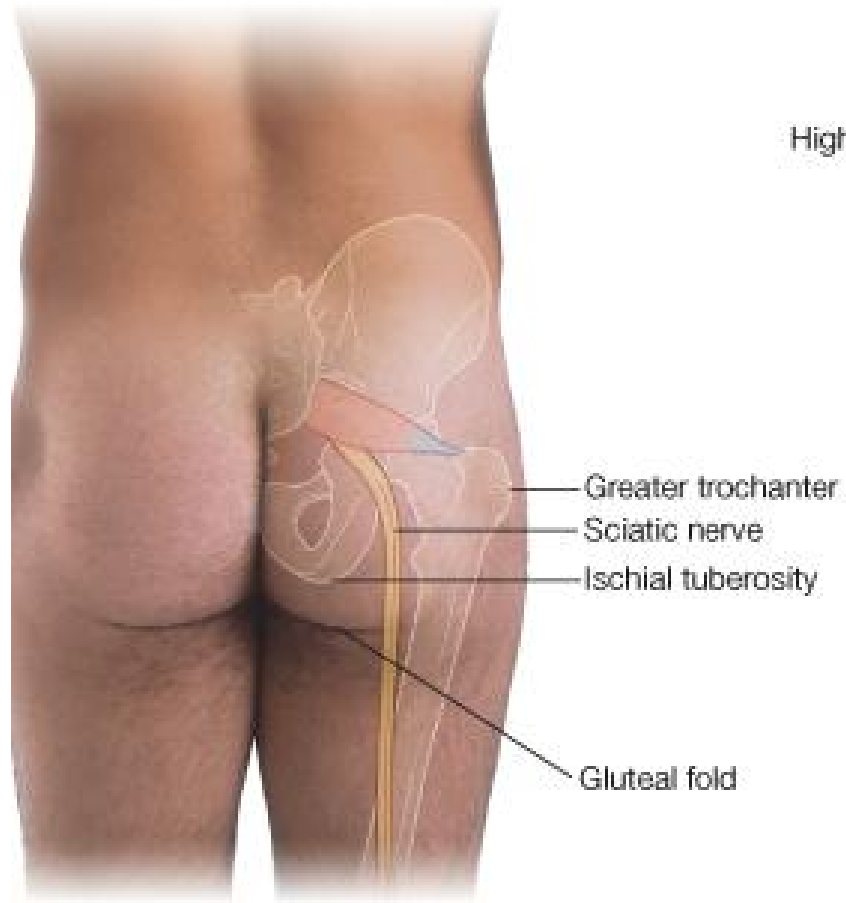
- derived from the deep peroneal and tibial nerves

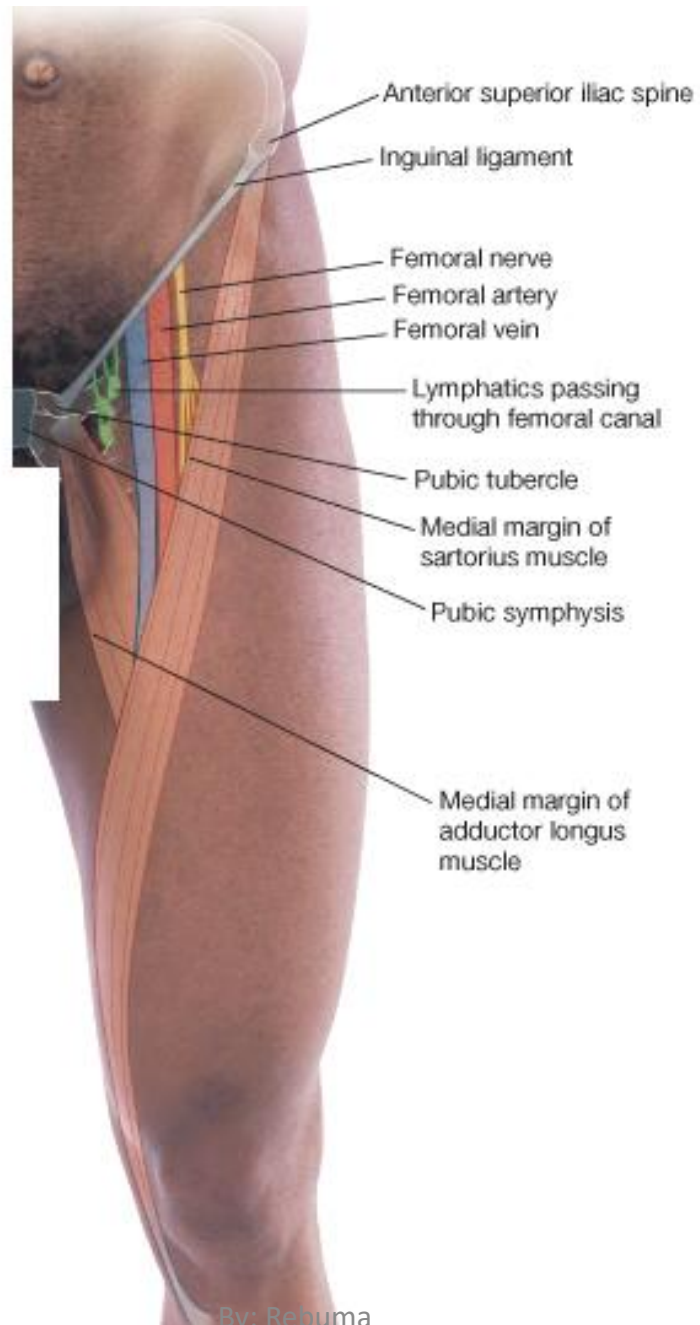


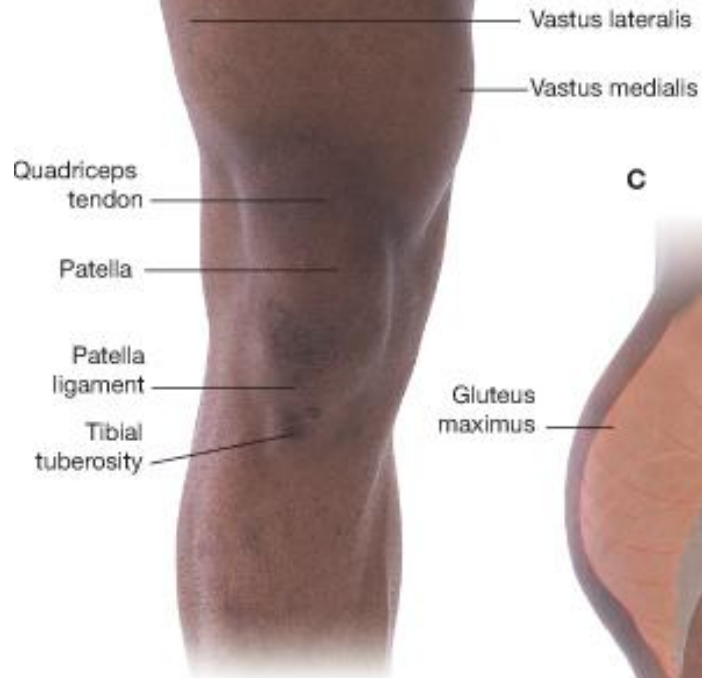
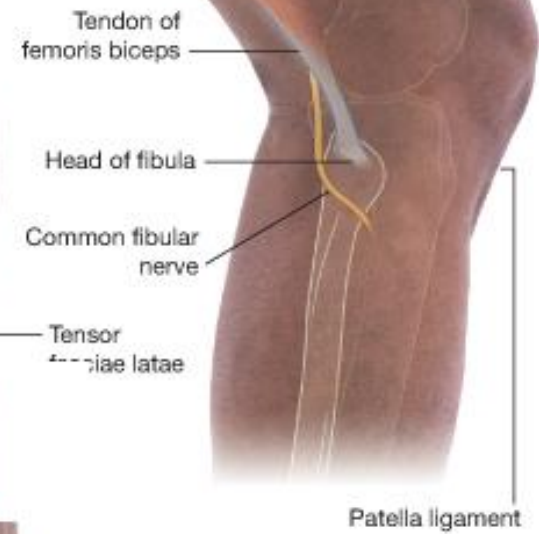


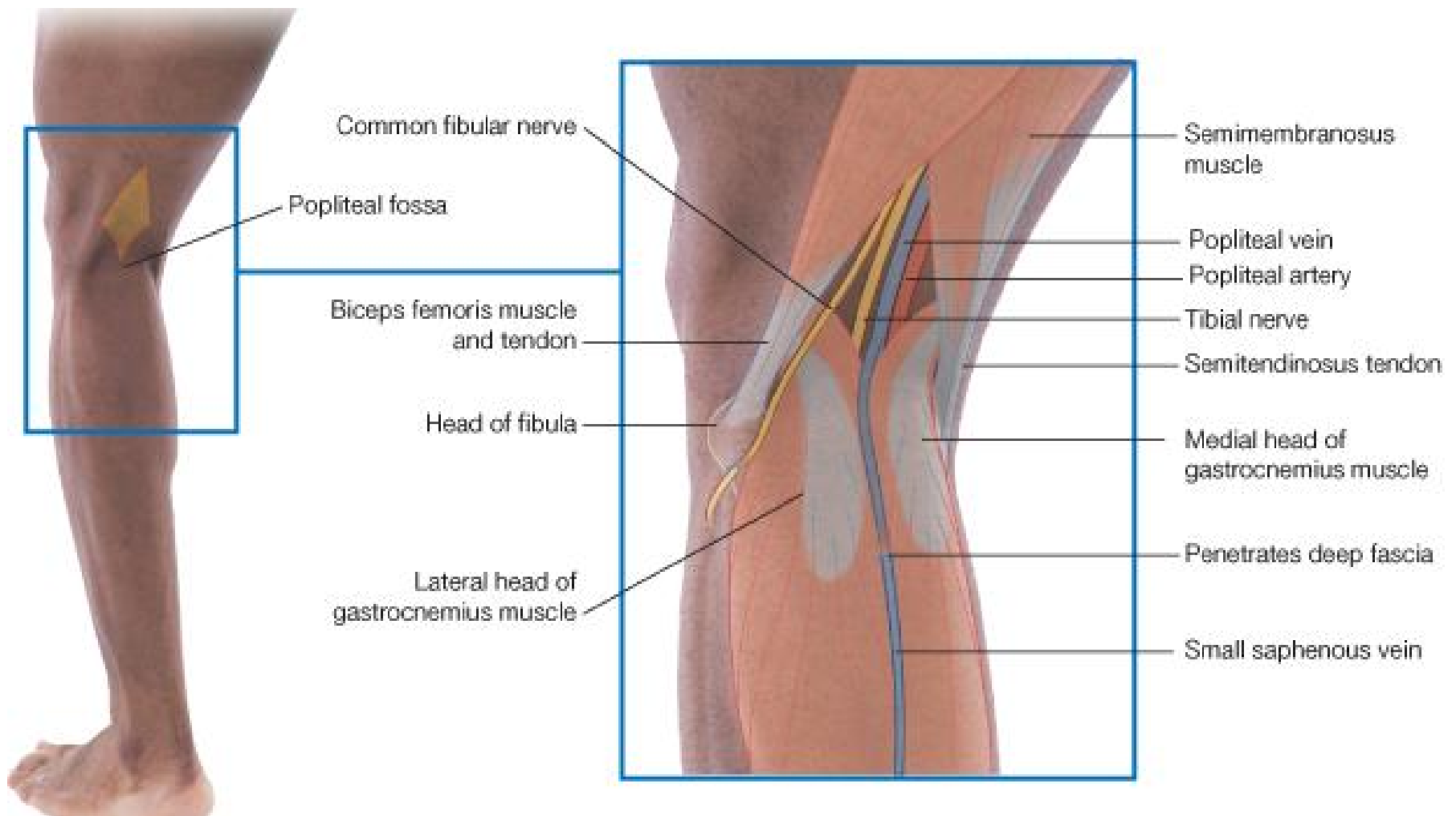
Peripheral pulses in the lower limb

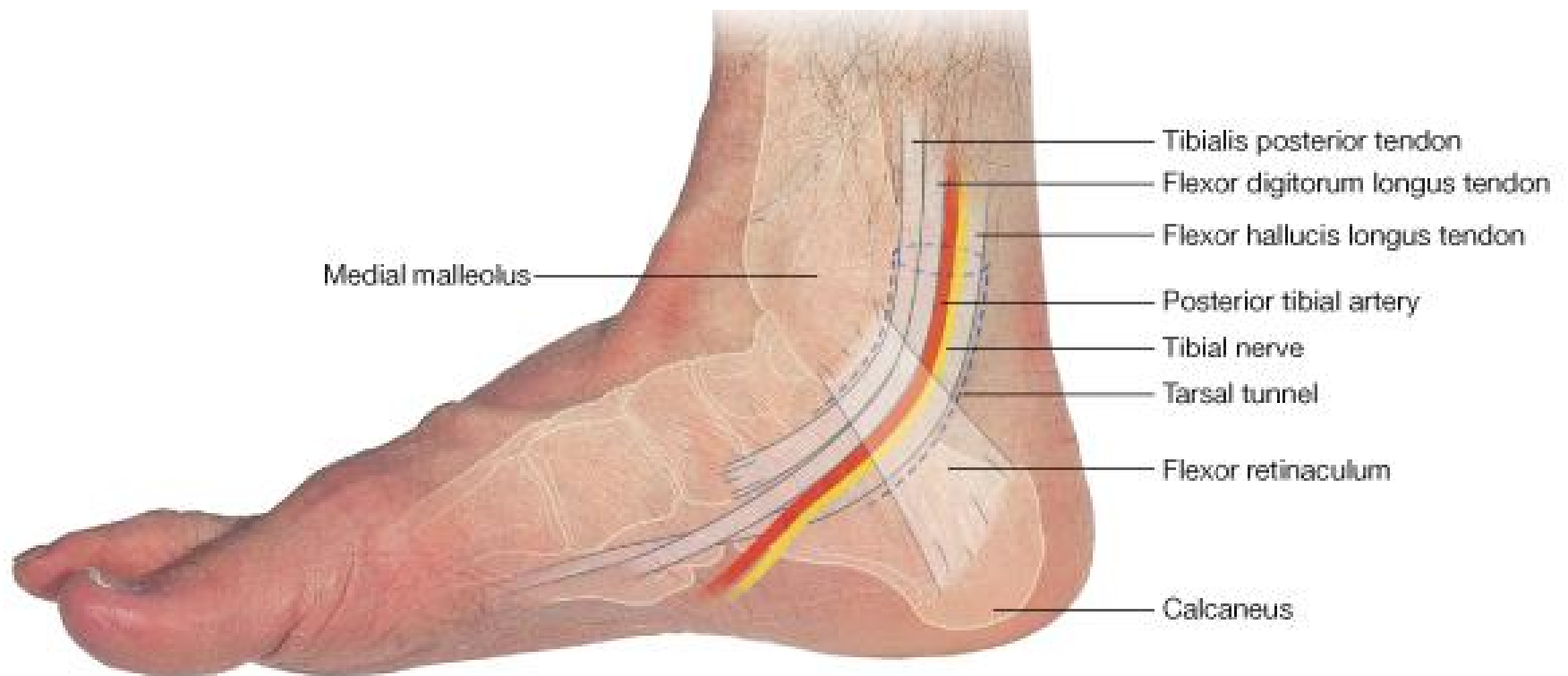
- Femoral pulse in the femoral triangle-femoral artery
- Popliteal pulse in the popliteal fossa-popliteal artery
- Posterior tibial pulse in the 'tarsal tunnel'-posterior tibial artery
- Dorsalis pedis pulse on the dorsal aspect of the foot-dorsalis pedis artery

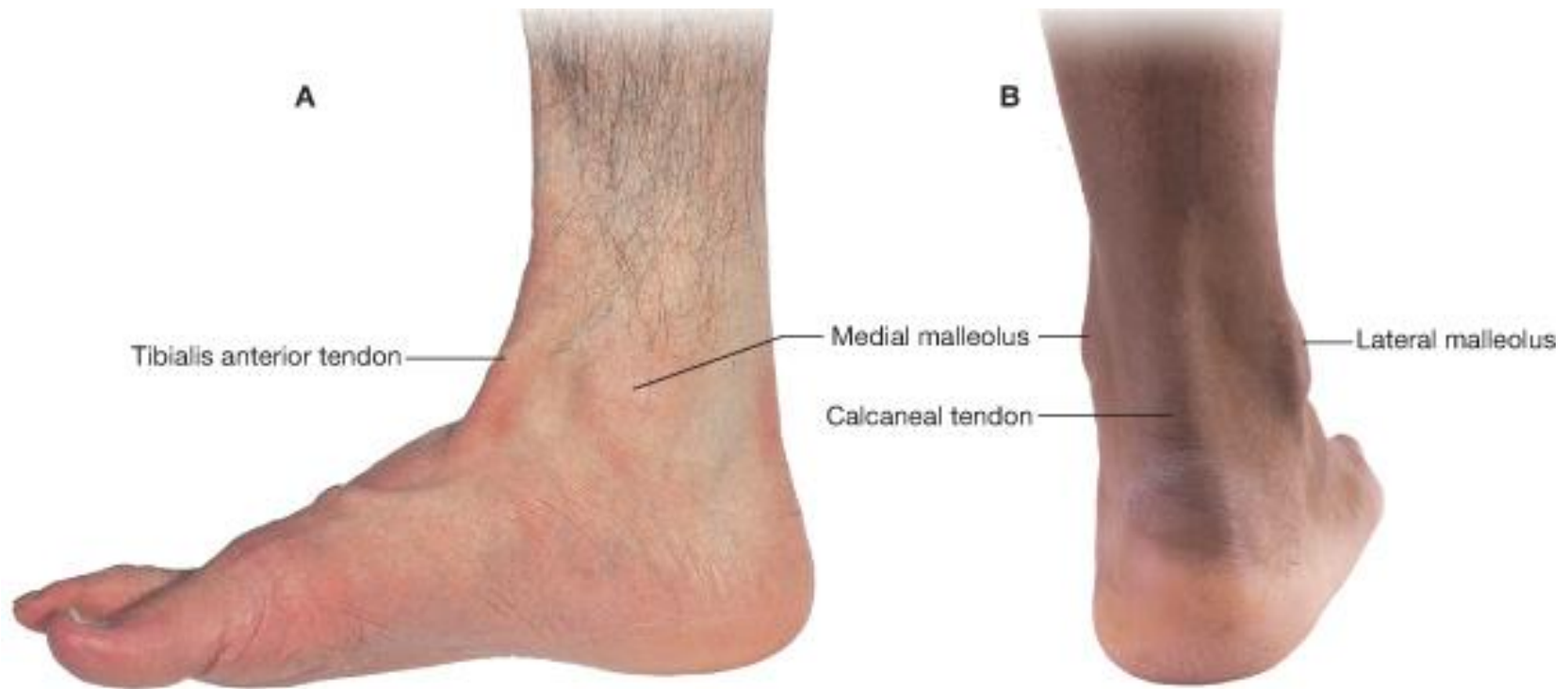




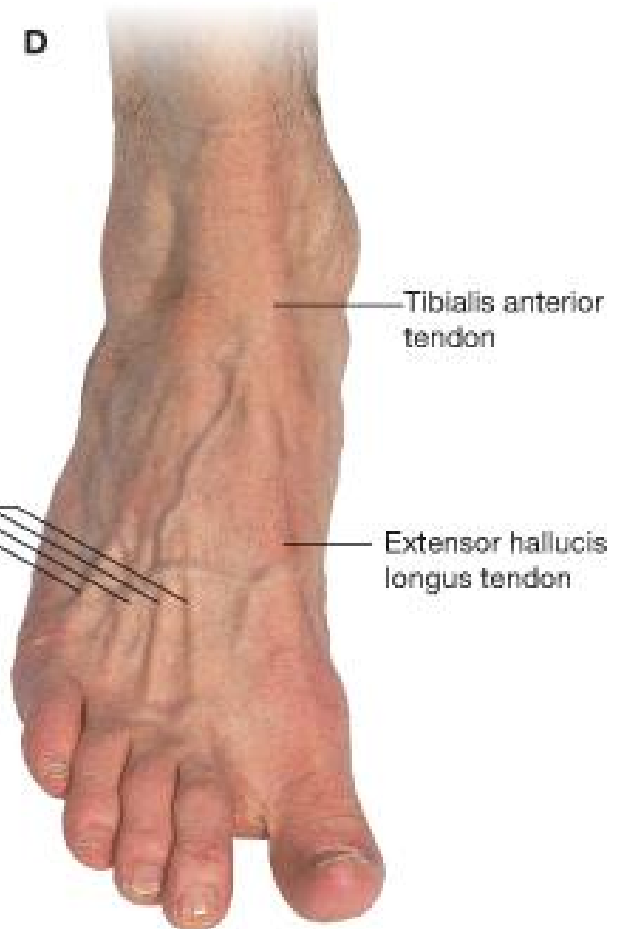
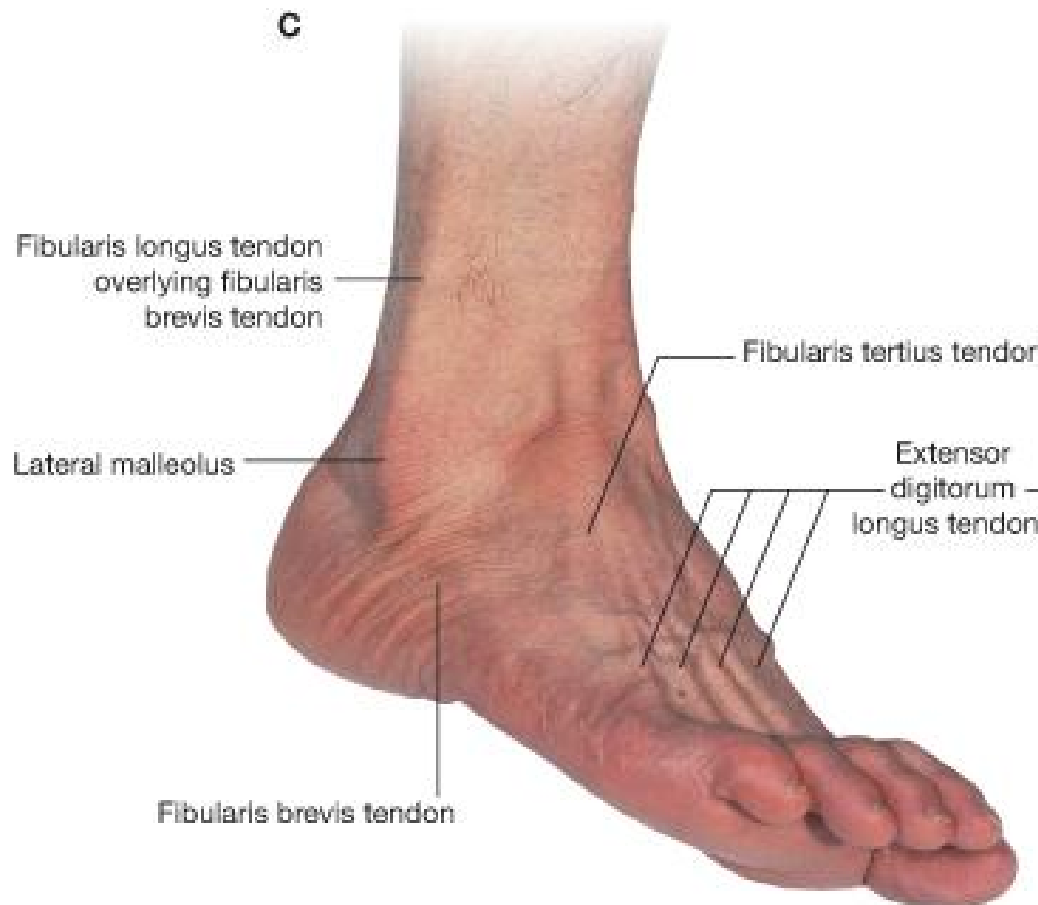
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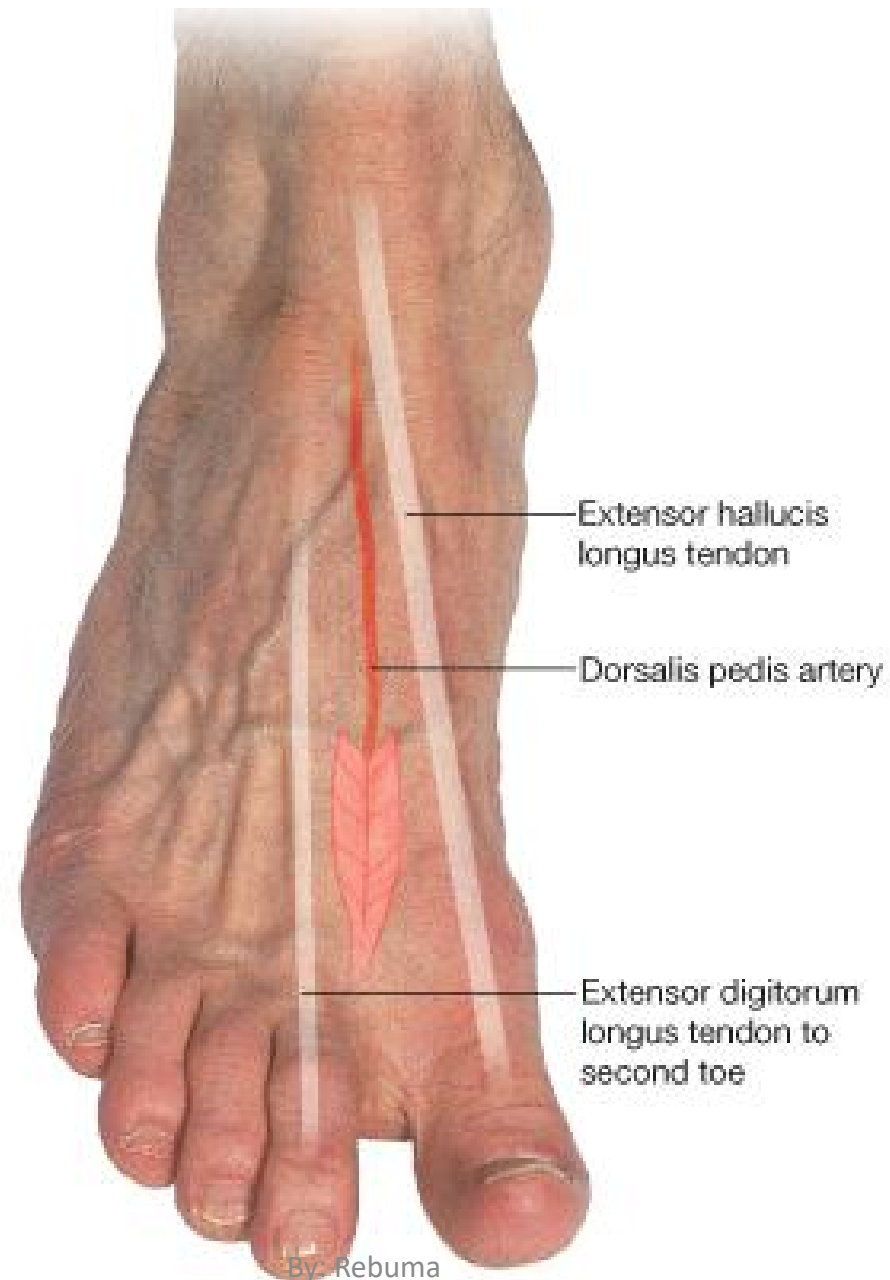


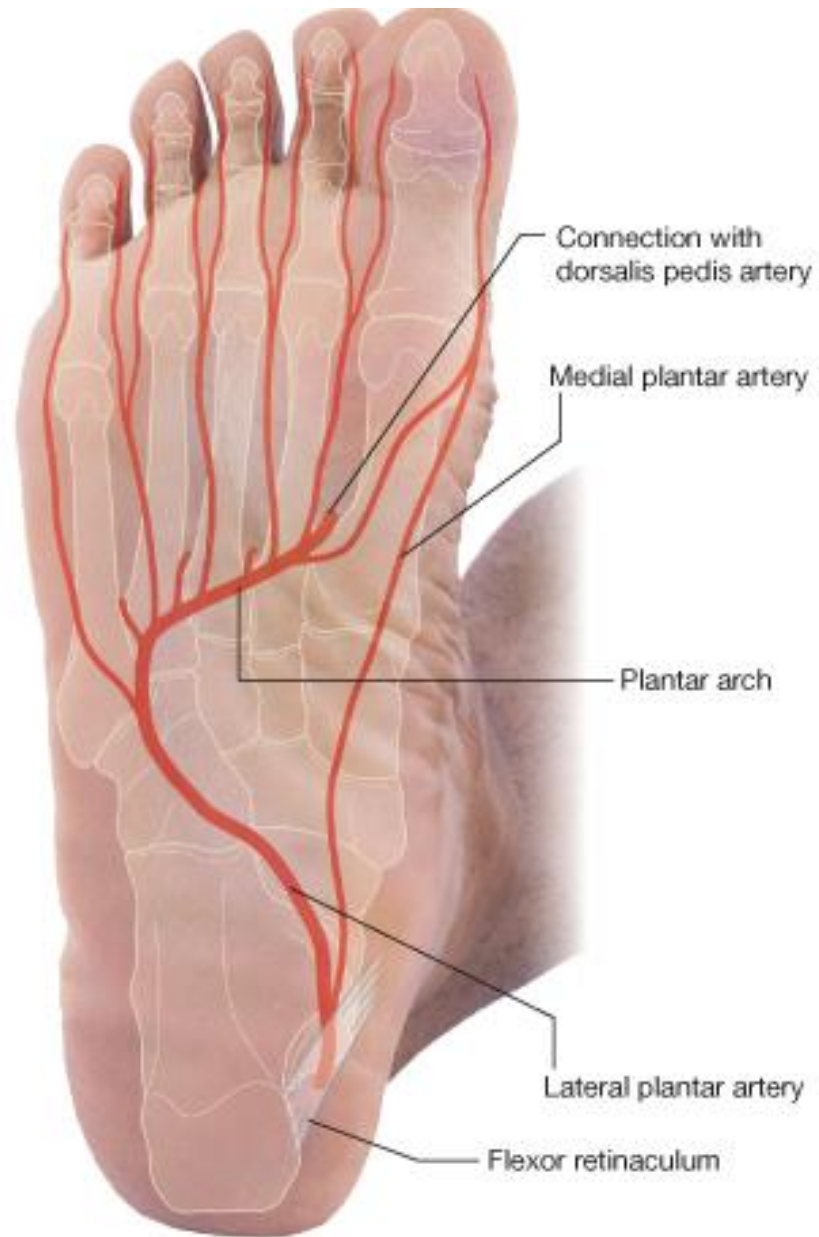




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To go far go together, to go fast go alone!

THANK YOU!